

**Department of Computer Science UET,
Lahore (New Campus), KSK**



Course: Computer Networks

Submitted By

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OEL FORM

1. You are designing a network for:

The network is designed for a multi-campus deployment of the **University of Engineering and Technology (UET) Lahore**.

2. No of Separate Units in above mentioned:

The design includes **three separate units**: the Main Campus Hub, Sub-Campus 1, and Sub-Campus 2, each operating on an isolated IP subnet.

3. Define the cost or budget of the above mentioned in terms of networking devices: The hardware inventory consists of **4 Internal Routers, 4 Border Routers, 1 ISP Router, 3 Layer 3 Core Switches, and 6 Layer 2 Access Switches**.

4. Define the number of End Devices allowed in each Unit:

Each unit accommodates **30 Personal Computers, 4 Wireless Devices 2 Tabs and Two Smart Phones, 1 printer for each department and 30 Cellular Devices** connected via a localized Wireless LAN Controller deployment.

5. Define the number of Border Router:

The topology incorporates exactly **4 Border Routers** to create a highly redundant, fault tolerant WAN core mesh infrastructure.

6. Define the number of Routers connected to ISP for internet:

Exactly 1 designated Border Router at the Main Campus edge connects the university infrastructure directly to the service provider.

7. Communication amongst the Units:

All **three campus units maintain complete, unrestricted inter-communication** across the corporate WAN backbone using optimized routing paths.

8. Define the number of servers and Types:

The data center houses **3 centralized enterprise servers**: a dedicated HTTP Web Server, an SMTP/POP3 Email Server, and a secure FTP Storage Server

9. Un-authorized Access:

An **Extended Access Control List (ACL)** isolates the Sub-Campus 1 Admin subnet from accessing the central HTTP Web Server over Port 80, throwing a request timeout.

10. Routing Protocols used:

The infrastructure utilizes **OSPF Area 0 for all dynamic routing between internal nodes**, and a **Static Default Route to connect upstream with the ISP**.

11. Private IP used (select one prefix):

The private network footprint utilizes the **192.168.0.0/16 prefix block**, using dynamic DHCP pools for the sub-campus and static addressing for the Main Campus servers.

12. No. of Public Addresses allocated by ISP to above mentioned:

The ISP has allocated **10 public IP addresses (105.12.32.15 – 24/25)**, reserving the last two for static server mapping and the rest for NAT Overload (PAT).

1. Abstract

This report documents the complete design and implementation of a hierarchical multi-campus university network using Cisco Packet Tracer. The network consists of one Main Campus and two Sub-Campuses, each with three departments (CS, EE, Administration) and wireless zones. The architecture deploys 8 routers (4 border routers including ISP, 3 internal routers, and 1 server farm router), 3 Layer 3 switches for inter-VLAN routing, and 3 Layer 2 access switches. Routing is handled by OSPF internally and BGP between campus border routers and the ISP. A centralized server farm at the Main Campus hosts Web (192.168.1.5), Email (192.168.1.6 – university.com), and FTP (192.168.1.7) services. Extended ACLs enforce role-based security, blocking Admin departments from HTTP while permitting email access.

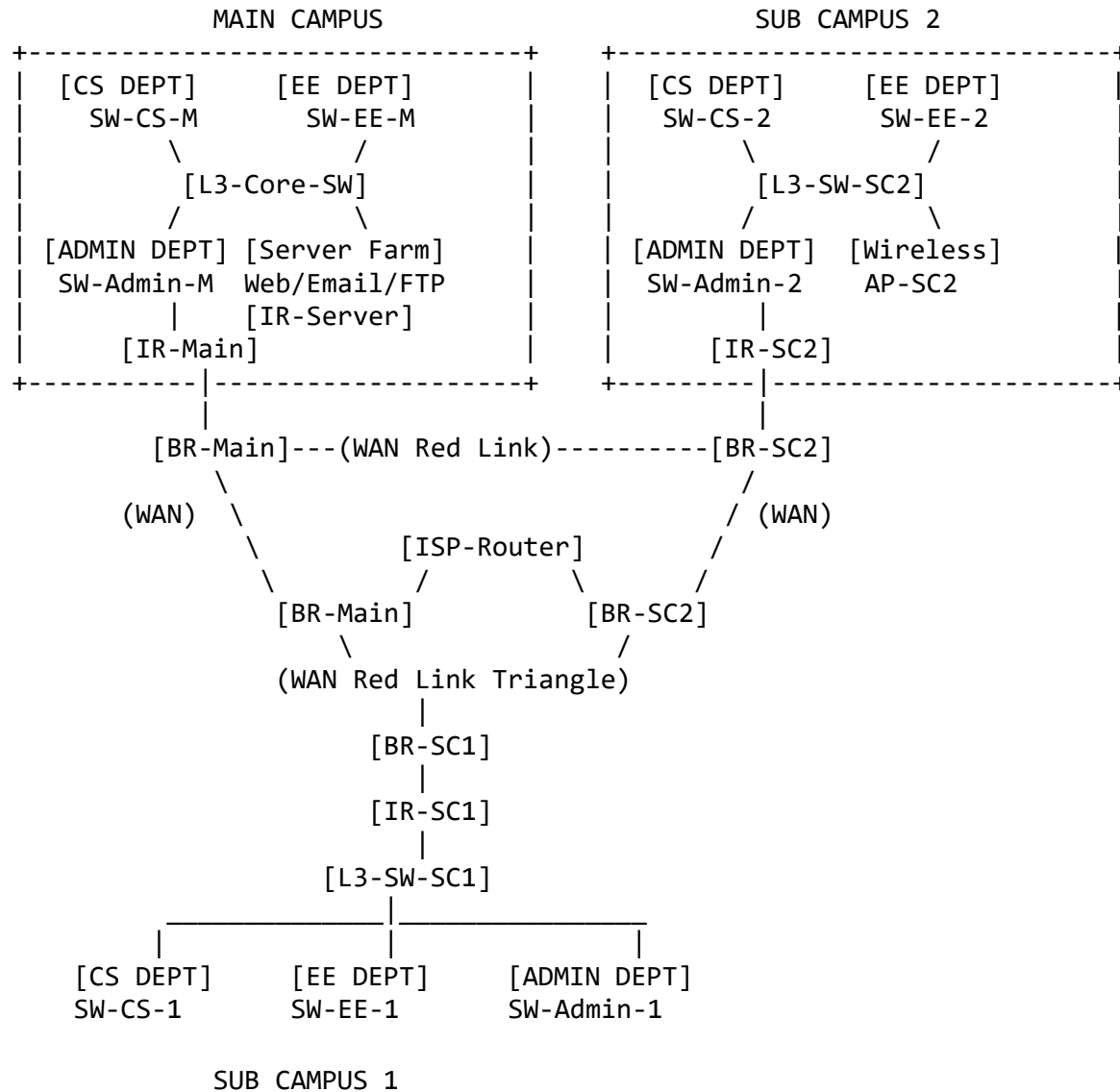
2. Problem Statement

Requirements satisfied by this implementation:

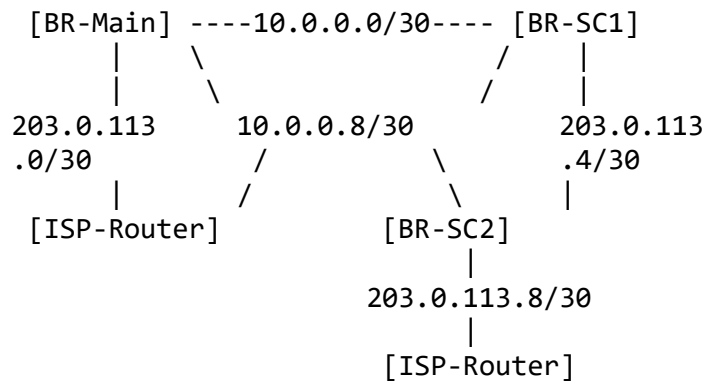
- 1 Main Campus + 2 Sub-Campuses with 3 departments each
- 8 Routers: 4 Border (BR-Main, BR-SC1, BR-SC2, ISP-Router) + 3 Internal (IR-Main, IR-SC1, IR-SC2) + 1 Server Farm (IR-Server)
- 3 Layer 3 Switches for Inter-VLAN routing
- 3 Layer 2 Access Switches
- VLAN isolation per department at each campus
- DHCP for dynamic IP assignment
- Centralized Server Farm: Web, Email, FTP
- OSPF for internal dynamic routing
- BGP for border/ISP routing
- Extended ACLs for security enforcement
- Wireless subnets at each campus
- Redundant WAN triangle for OSPF failover

3. Network Topology Diagram

3.1 Full Topology Overview



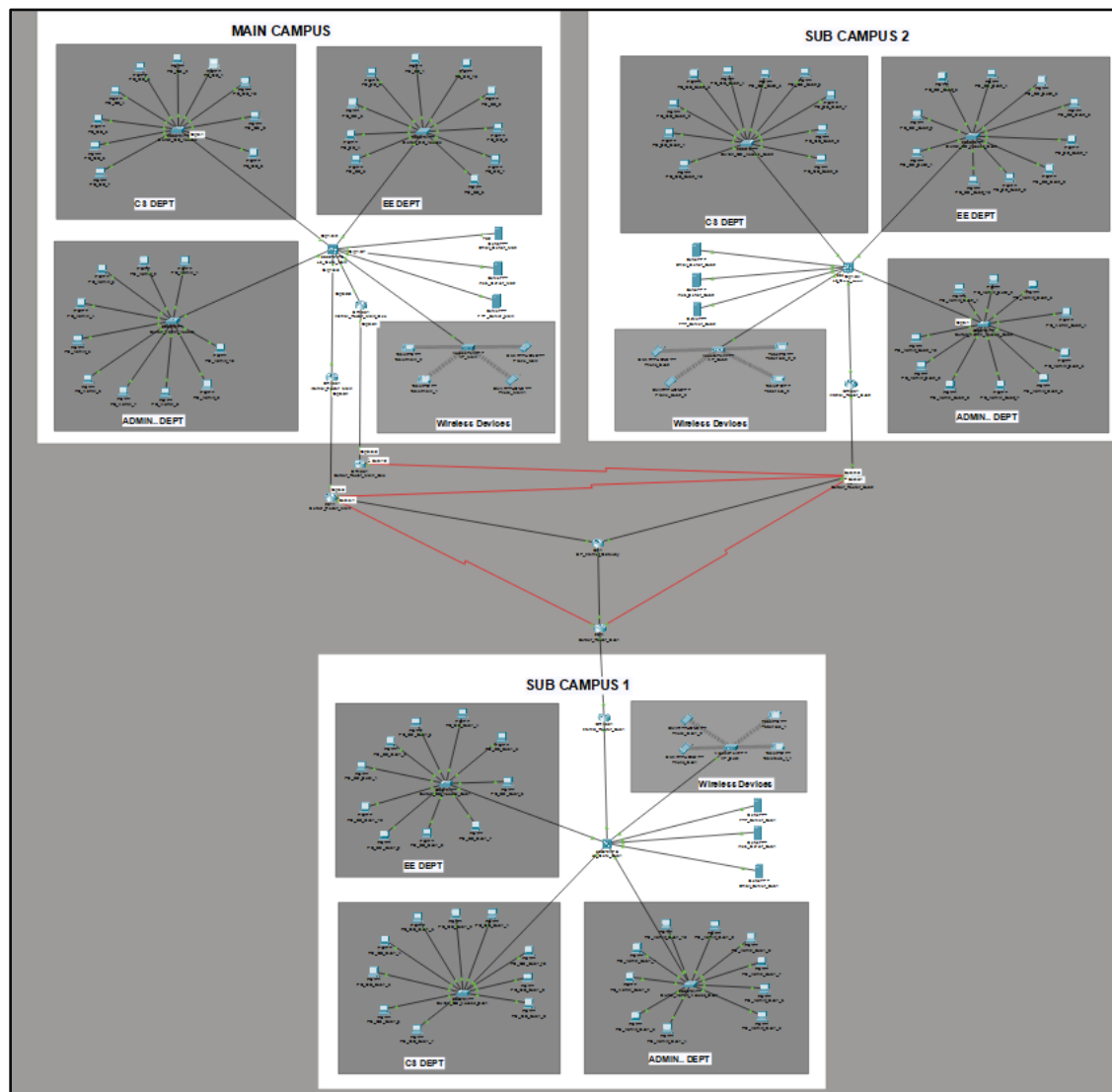
3.2 WAN Triangle Topology



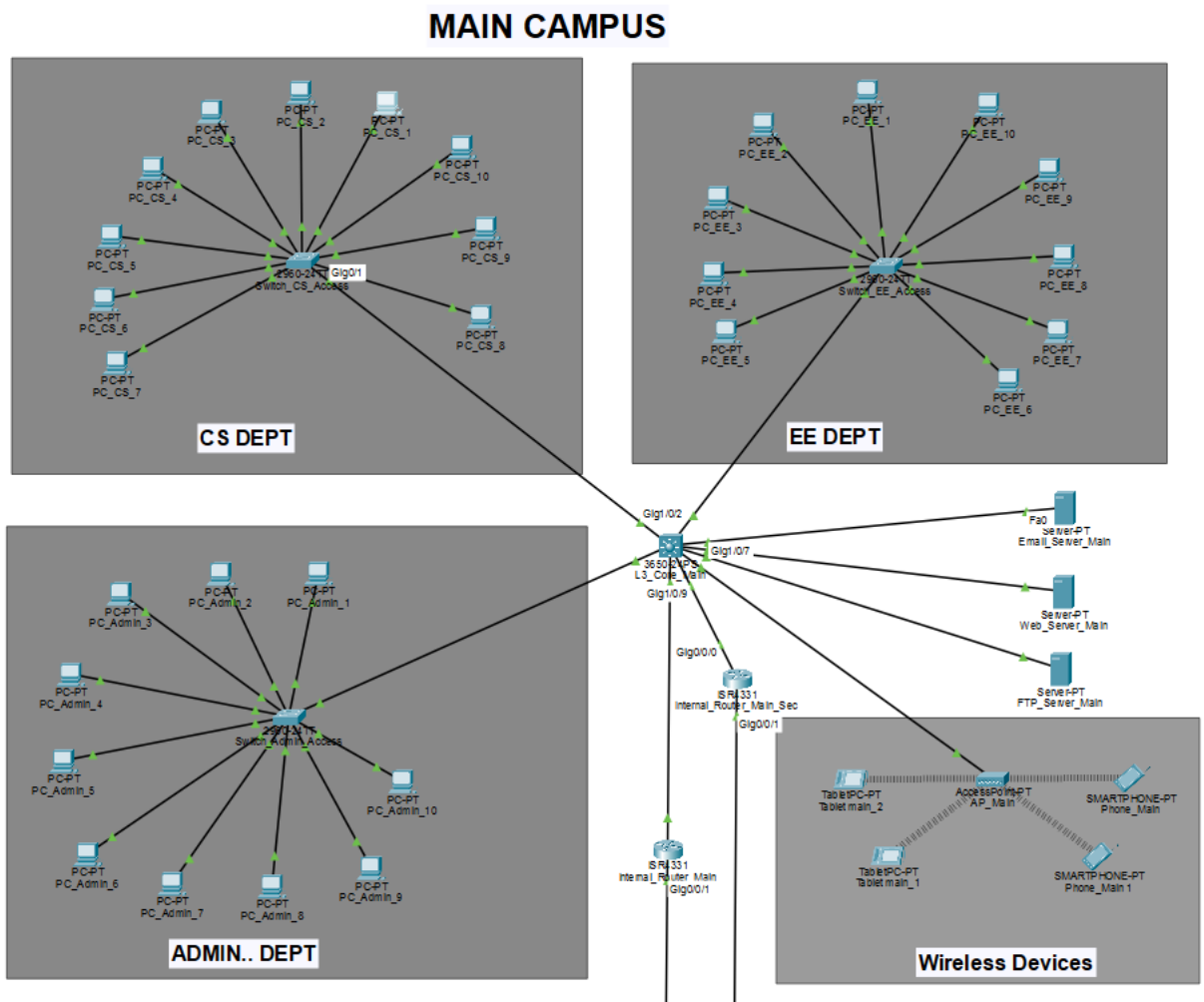
3.3 Router Summary

#	Router Name	Type	Location	BGP AS
1	BR-Main	Border Router	Main Campus	AS 65000
2	BR-SC1	Border Router	Sub-Campus 1	AS 65001
3	BR-SC2	Border Router	Sub-Campus 2	AS 65002
4	ISP-Router	ISP/Core Router	ISP Cloud	AS 65000
5	IR-Main	Internal Router	Main Campus	OSPF only
6	IR-SC1	Internal Router	Sub-Campus 1	OSPF only
7	IR-SC2	Internal Router	Sub-Campus 2	OSPF only
8	IR-Server	Server Farm Router	Main Campus	OSPF only

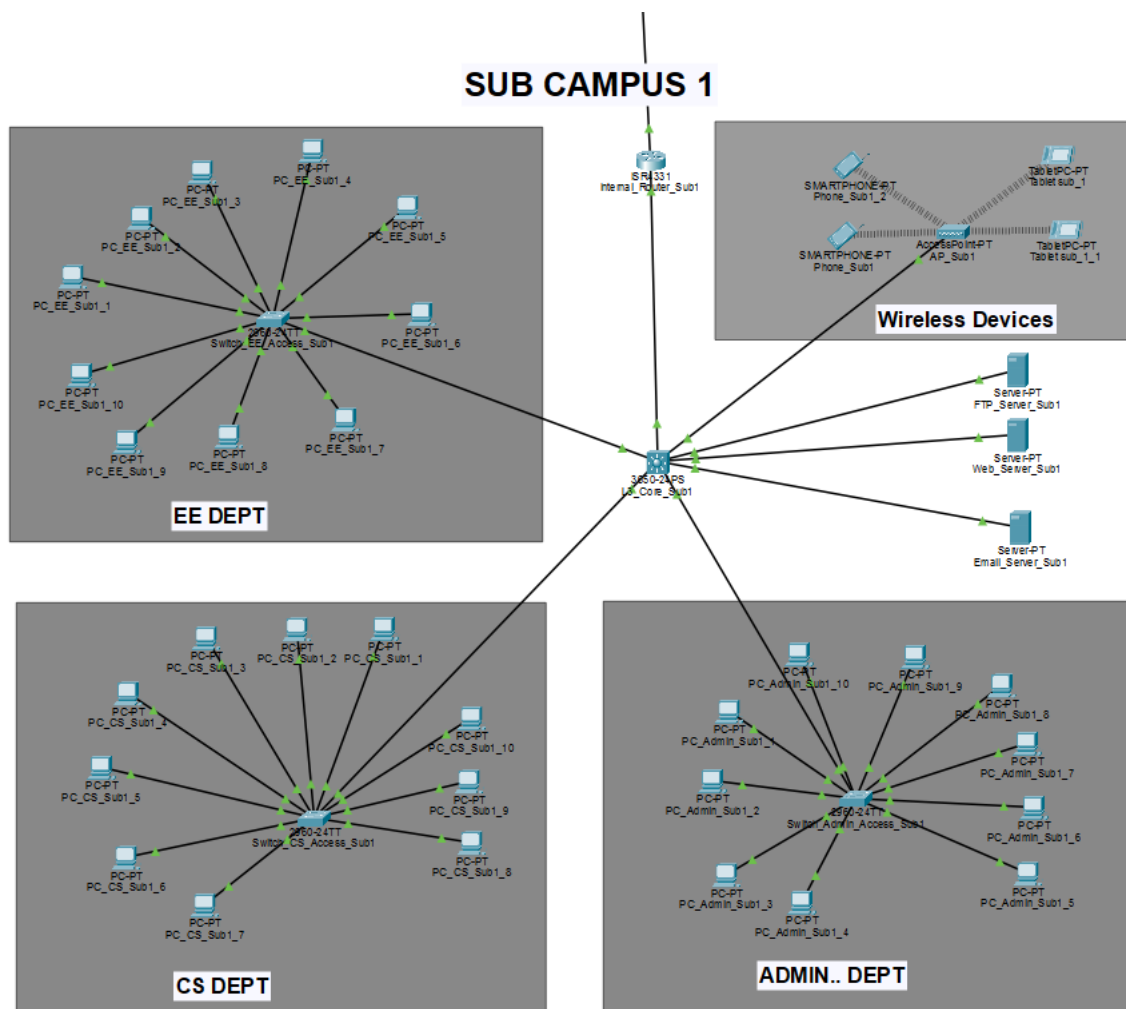
3.4 Full Topology



3.5 Main Campus Topology

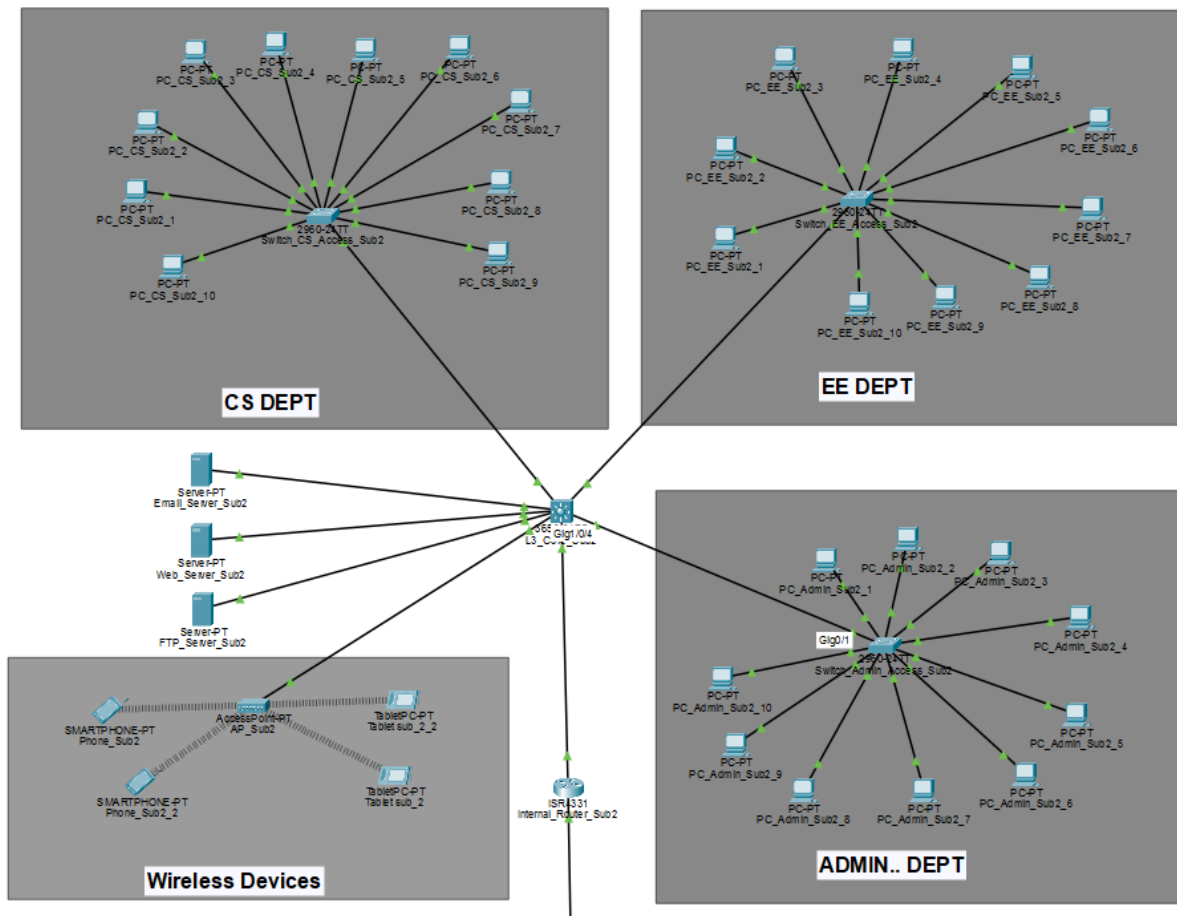


3.6 Sub Campus 1 Topology

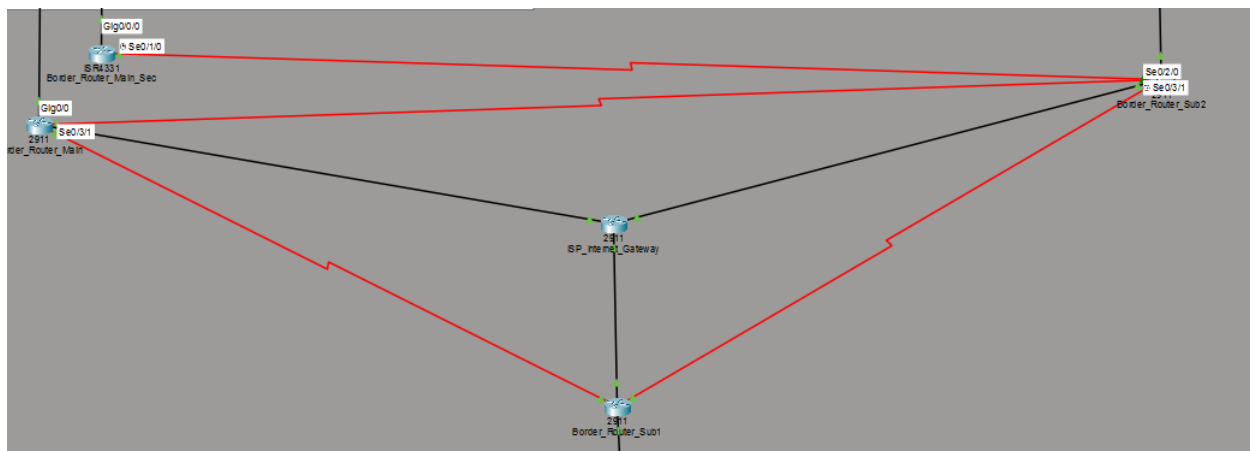


3.7 Sub Campus 2 Topology

SUB CAMPUS 2



3.8 WAN TRIANGLE / Border Routers Topology



4. IP Addressing & VLSM Details

All departments use /27 subnets (30 usable hosts). Private space 192.168.X.0/24 per campus.

4.1 Main Campus – 192.168.1.0/24

Department	VLAN	Network	Mask	Gateway	Host Range	Broadcast
Computer Science	VLAN 10	192.168.1.0/27	255.255.255.224	192.168.1.1	.11 – .20	192.168.1.31
Electrical Eng.	VLAN 20	192.168.1.32/27	255.255.255.224	192.168.1.33	.41 – .50	192.168.1.63
Administration	VLAN 30	192.168.1.64/27	255.255.255.224	192.168.1.65	.75 – .84	192.168.1.95
Server Farm	VLAN 50	192.168.1.0/27	255.255.255.224	192.168.1.1	.2 – .10	192.168.1.31
Wireless	VLAN 40	192.168.1.96/27	255.255.255.224	192.168.1.97	.98 – .126	192.168.1.127

4.2 Sub-Campus 1 – 192.168.2.0/24

Department	VLAN	Network	Mask	Gateway	Host Range	Broadcast
Computer Science	VLAN 10	192.168.2.0/27	255.255.255.224	192.168.2.1	.11 – .20	192.168.2.31
Electrical Eng.	VLAN 20	192.168.2.32/27	255.255.255.224	192.168.2.33	.41 – .50	192.168.2.63
Administration	VLAN 30	192.168.2.64/27	255.255.255.224	192.168.2.65	.71 – .80	192.168.2.95
Wireless	VLAN 41	192.168.2.96/27	255.255.255.224	192.168.2.97	.98 – .126	192.168.2.127

4.3 Sub-Campus 2 – 192.168.3.0/24

Department	VLAN	Network	Mask	Gateway	Host Range	Broadcast
Computer Science	VLAN 10	192.168.3.0/27	255.255.255.224	192.168.3.1	.11 – .20	192.168.3.31
Electrical Eng.	VLAN 20	192.168.3.32/27	255.255.255.224	192.168.3.33	.41 – .50	192.168.3.63
Administration	VLAN 30	192.168.3.64/27	255.255.255.224	192.168.3.65	.71 – .80	192.168.3.95
Wireless	VLAN 42	192.168.3.96/27	255.255.255.224	192.168.3.97	.98 – .126	192.168.3.127

4.4 Server Farm – Static IPs (Main Campus)

Server	IP Address	Mask	Gateway	Services
Web_Server_Main	192.168.1.5	255.255.255.224	192.168.1.1	HTTP (80), HTTPS (443)
Email_Server_Main	192.168.1.6	255.255.255.224	192.168.1.1	SMTP (25), POP3 (110)
FTP_Server_Main	192.168.1.7	255.255.255.224	192.168.1.1	FTP (21)

4.5 WAN Point-to-Point Links (/30)

Link	Network	Router A	IP A	Router B	IP B
Main <-> ISP	203.0.113.0/30	BR-Main	203.0.113.1	ISP-Router	203.0.113.2
SC1 <-> ISP	203.0.113.4/30	BR-SC1	203.0.113.5	ISP-Router	203.0.113.6
SC2 <-> ISP	203.0.113.8/30	BR-SC2	203.0.113.9	ISP-Router	203.0.113.10
Main <-> SC1	10.0.0.0/30	BR-Main	10.0.0.1	BR-SC1	10.0.0.2
Main <-> SC2	10.0.0.4/30	BR-Main	10.0.0.5	BR-SC2	10.0.0.6
SC1 <-> SC2	10.0.0.8/30	BR-SC1	10.0.0.9	BR-SC2	10.0.0.10

4.6 Internal Router Links (/30)

Link	Network	Router A	IP A	Device B	IP B
IR-Main <-> L3-Core-SW	10.1.0.0/30	IR-Main	10.1.0.1	L3-Core-SW	10.1.0.2
IR-Main <-> BR-Main	10.1.0.4/30	IR-Main	10.1.0.5	BR-Main	10.1.0.6
IR-Server <-> L3-Core-SW	10.1.0.8/30	IR-Server	10.1.0.9	L3-Core-SW	10.1.0.10
IR-SC1 <-> L3-SW-SC1	10.2.0.0/30	IR-SC1	10.2.0.1	L3-SW-SC1	10.2.0.2
IR-SC1 <-> BR-SC1	10.2.0.4/30	IR-SC1	10.2.0.5	BR-SC1	10.2.0.6
IR-SC2 <-> L3-SW-SC2	10.3.0.0/30	IR-SC2	10.3.0.1	L3-SW-SC2	10.3.0.2
IR-SC2 <-> BR-SC2	10.3.0.4/30	IR-SC2	10.3.0.5	BR-SC2	10.3.0.6

5. VLAN Planning Table

VLAN ID	Name	Department	Campus
VLAN 10	CS_DEPT	Computer Science	All Campuses
VLAN 20	EE_DEPT	Electrical Engineering	All Campuses
VLAN 30	ADMIN_DEPT	Administration	All Campuses
VLAN 40	WIRELESS_MAIN	Wireless Users	Main Campus
VLAN 41	WIRELESS_SC1	Wireless Users	Sub-Campus 1
VLAN 42	WIRELESS_SC2	Wireless Users	Sub-Campus 2
VLAN 50	SERVER_FARM	Servers	Main Campus
VLAN 99	MGMT	Management	All Campuses

6. Device Inventory

Device	Type	Location	Primary IP	Role
BR-Main	Router 2911	Main Campus	203.0.113.1	Border Router (BGP AS65000)
BR-SC1	Router 2911	Sub-Campus 1	203.0.113.5	Border Router (BGP AS65001)
BR-SC2	Router 2911	Sub-Campus 2	203.0.113.9	Border Router (BGP AS65002)
ISP-Router	Router 2911	ISP Cloud	203.0.113.2	ISP / Core Router
IR-Main	Router 2911	Main Campus	10.1.0.1	Internal Router OSPF
IR-SC1	Router 2911	Sub-Campus 1	10.2.0.1	Internal Router OSPF
IR-SC2	Router 2911	Sub-Campus 2	10.3.0.1	Internal Router OSPF
IR-Server	Router 2911	Main Campus	10.1.0.9	Server Farm Router OSPF
L3-Core-SW	Switch 3560	Main Campus	SVI mgmt	Inter-VLAN + DHCP
L3-SW-SC1	Switch 3560	Sub-Campus 1	SVI mgmt	Inter-VLAN + DHCP
L3-SW-SC2	Switch 3560	Sub-Campus 2	SVI mgmt	Inter-VLAN + DHCP
SW-Main	Switch 2960	Main Campus	–	Access Switch
SW-SC1	Switch 2960	Sub-Campus 1	–	Access Switch
SW-SC2	Switch 2960	Sub-Campus 2	–	Access Switch
Web_Server_Main	Server	Main Campus	192.168.1.5	HTTP/HTTPS
Email_Server_Main	Server	Main Campus	192.168.1.6	SMTP/POP3
FTP_Server_Main	Server	Main Campus	192.168.1.7	FTP
AP-Main	Access Point	Main Campus	DHCP	Wireless VLAN 40
AP-SC1	Access Point	Sub-Campus 1	DHCP	Wireless VLAN 41
AP-SC2	Access Point	Sub-Campus 2	DHCP	Wireless VLAN 42

7. Router Configuration – All 8 Routers

Router 1: ISP-Router (Central ISP – AS 65000)

This router simulates the Internet Service Provider. It peers with all three campus border routers via BGP.

```
Router> enable
Router# configure terminal
Router(config)# hostname ISP-Router
```

```
! Interface to BR-Main
ISP-Router(config)# interface GigabitEthernet0/0
ISP-Router(config-if)# ip address 203.0.113.2 255.255.255.252
ISP-Router(config-if)# no shutdown
ISP-Router(config-if)# exit
```

```
! Interface to BR-SC1
ISP-Router(config)# interface GigabitEthernet0/1
ISP-Router(config-if)# ip address 203.0.113.6 255.255.255.252
ISP-Router(config-if)# no shutdown
ISP-Router(config-if)# exit
```

```
! Interface to BR-SC2
ISP-Router(config)# interface GigabitEthernet0/2
ISP-Router(config-if)# ip address 203.0.113.10 255.255.255.252
ISP-Router(config-if)# no shutdown
ISP-Router(config-if)# exit
```

```
! BGP Configuration -- AS 65000 (ISP)
ISP-Router(config)# router bgp 65000
ISP-Router(config-router)# bgp router-id 9.9.9.9
ISP-Router(config-router)# neighbor 203.0.113.1 remote-as 65000
ISP-Router(config-router)# neighbor 203.0.113.5 remote-as 65001
ISP-Router(config-router)# neighbor 203.0.113.9 remote-as 65002
ISP-Router(config-router)# network 203.0.113.0 mask 255.255.255.252
ISP-Router(config-router)# network 203.0.113.4 mask 255.255.255.252
ISP-Router(config-router)# network 203.0.113.8 mask 255.255.255.252
ISP-Router(config-router)# exit
```

```
ISP-Router(config)# end
ISP-Router# write memory
```

ISP-Router

IP BGP summary

```
Router#show ip bgp summary
BGP router identifier 105.12.32.17, local AS number 65000
BGP table version is 1, main routing table version 6
0 network entries using 0 bytes of memory
0 path entries using 0 bytes of memory
0/0 BGP path/bestpath attribute entries using 0 bytes of memory
0 BGP AS-PATH entries using 0 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 32 total bytes of memory
BGP activity 0/0 prefixes, 0/0 paths, scan interval 60 secs
```

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
203.0.113.1	4	100	0	0	1	0	0	07:14:37	4
203.0.113.5	4	200	0	0	1	0	0	07:14:37	4
203.0.113.9	4	300	0	0	1	0	0	07:14:37	4
203.0.113.13	4	100	0	0	1	0	0	07:14:37	4

IP Route

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

105.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C       105.12.32.0/29 is directly connected, GigabitEthernet0/0
L       105.12.32.1/32 is directly connected, GigabitEthernet0/0
C       105.12.32.8/29 is directly connected, GigabitEthernet0/1
L       105.12.32.9/32 is directly connected, GigabitEthernet0/1
C       105.12.32.16/29 is directly connected, GigabitEthernet0/2
L       105.12.32.17/32 is directly connected, GigabitEthernet0/2
```

Secondary Border Router – Main

IP BGP summary

```

Border_Router_Main_Sec#show ip bgp summary
BGP router identifier 203.0.113.13, local AS number 100
BGP table version is 1, main routing table version 6
0 network entries using 0 bytes of memory
0 path entries using 0 bytes of memory
0/0 BGP path/bestpath attribute entries using 0 bytes of memory
0 BGP AS-PATH entries using 0 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 32 total bytes of memory
BGP activity 0/0 prefixes, 0/0 paths, scan interval 60 secs

Neighbor      V    AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
203.0.113.14   4 65000      0       0        1    0    0 01:34:14      4

```

IP Route

```

Border_Router_Main_Sec#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is 10.0.0.14 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
    C       10.0.0.12/30 is directly connected, Serial0/1/0
    L       10.0.0.13/32 is directly connected, Serial0/1/0
    C       10.10.15.0/30 is directly connected, GigabitEthernet0/0/0
    L       10.10.15.2/32 is directly connected, GigabitEthernet0/0/0
    192.168.1.0/27 is subnetted, 4 subnets
    O       192.168.1.0/27 [110/131] via 10.0.0.14, 00:10:10, Serial0/1/0
    O       192.168.1.32/27 [110/131] via 10.0.0.14, 00:10:10, Serial0/1/0
    O       192.168.1.64/27 [110/131] via 10.0.0.14, 00:10:10, Serial0/1/0
    O       192.168.1.96/27 [110/131] via 10.0.0.14, 00:10:10, Serial0/1/0
    192.168.2.0/27 is subnetted, 4 subnets
    O       192.168.2.0/27 [110/131] via 10.0.0.14, 00:31:38, Serial0/1/0
    O       192.168.2.32/27 [110/131] via 10.0.0.14, 00:31:38, Serial0/1/0
    O       192.168.2.64/27 [110/131] via 10.0.0.14, 00:31:38, Serial0/1/0
    O       192.168.2.96/27 [110/131] via 10.0.0.14, 00:31:38, Serial0/1/0
    192.168.3.0/27 is subnetted, 4 subnets
    O       192.168.3.0/27 [110/67] via 10.0.0.14, 00:31:38, Serial0/1/0
    O       192.168.3.32/27 [110/67] via 10.0.0.14, 00:31:38, Serial0/1/0
    O       192.168.3.64/27 [110/67] via 10.0.0.14, 00:31:38, Serial0/1/0
    O       192.168.3.96/27 [110/67] via 10.0.0.14, 00:31:38, Serial0/1/0
    192.168.10.0/30 is subnetted, 3 subnets
    O       192.168.10.0/30 [110/130] via 10.0.0.14, 00:31:38, Serial0/1/0
    O       192.168.10.4/30 [110/130] via 10.0.0.14, 00:31:38, Serial0/1/0
    O       192.168.10.8/30 [110/66] via 10.0.0.14, 00:31:38, Serial0/1/0
    192.168.20.0/30 is subnetted, 3 subnets
    O       192.168.20.0/30 [110/129] via 10.0.0.14, 00:31:38, Serial0/1/0
    O       192.168.20.4/30 [110/129] via 10.0.0.14, 00:31:38, Serial0/1/0
    O       192.168.20.8/30 [110/65] via 10.0.0.14, 00:31:38, Serial0/1/0
    192.168.30.0/30 is subnetted, 3 subnets
    O       192.168.30.0/30 [110/192] via 10.0.0.14, 00:31:38, Serial0/1/0
    O       192.168.30.4/30 [110/128] via 10.0.0.14, 00:31:38, Serial0/1/0
    O       192.168.30.8/30 [110/128] via 10.0.0.14, 00:31:38, Serial0/1/0
O*E2 0.0.0.0/0 [110/1] via 10.0.0.14, 00:31:38, Serial0/1/0

```

Router 2: BR-Main (Main Campus Border Router – AS 65000)

Connects Main Campus to ISP and peers directly with BR-SC1 and BR-SC2 via the WAN triangle.

```
Router> enable
Router# configure terminal
Router(config)# hostname BR-Main
```

```
! Interface to ISP-Router
BR-Main(config)# interface GigabitEthernet0/0
BR-Main(config-if)# ip address 203.0.113.1 255.255.255.252
BR-Main(config-if)# description Link-to-ISP
BR-Main(config-if)# no shutdown
BR-Main(config-if)# exit
```

```
! Interface to BR-SC1 (WAN Triangle)
BR-Main(config)# interface GigabitEthernet0/1
BR-Main(config-if)# ip address 10.0.0.1 255.255.255.252
BR-Main(config-if)# description WAN-Link-to-BR-SC1
BR-Main(config-if)# no shutdown
BR-Main(config-if)# exit
```

```
! Interface to BR-SC2 (WAN Triangle)
BR-Main(config)# interface GigabitEthernet0/2
BR-Main(config-if)# ip address 10.0.0.5 255.255.255.252
BR-Main(config-if)# description WAN-Link-to-BR-SC2
BR-Main(config-if)# no shutdown
BR-Main(config-if)# exit
```

```
! Interface to IR-Main (Internal)
BR-Main(config)# interface GigabitEthernet1/0
BR-Main(config-if)# ip address 10.1.0.6 255.255.255.252
BR-Main(config-if)# description Link-to-IR-Main
BR-Main(config-if)# no shutdown
BR-Main(config-if)# exit
```

```
! BGP Configuration -- AS 65000
BR-Main(config)# router bgp 65000
BR-Main(config-router)# bgp router-id 1.1.1.1
BR-Main(config-router)# neighbor 203.0.113.2 remote-as 65000
BR-Main(config-router)# neighbor 10.0.0.2 remote-as 65001
BR-Main(config-router)# neighbor 10.0.0.6 remote-as 65002
BR-Main(config-router)# network 192.168.1.0 mask 255.255.255.224
BR-Main(config-router)# network 192.168.1.32 mask 255.255.255.224
BR-Main(config-router)# network 192.168.1.64 mask 255.255.255.224
BR-Main(config-router)# network 192.168.1.96 mask 255.255.255.224
BR-Main(config-router)# exit
```

```

! OSPF -- redistribute connected routes
BR-Main(config)# router ospf 1
BR-Main(config-router)# router-id 1.1.1.1
BR-Main(config-router)# network 10.1.0.4 0.0.0.3 area 0
BR-Main(config-router)# network 10.0.0.0 0.0.0.3 area 0
BR-Main(config-router)# network 10.0.0.4 0.0.0.3 area 0
BR-Main(config-router)# default-information originate
BR-Main(config-router)# exit

```

```

! Default route to ISP
BR-Main(config)# ip route 0.0.0.0 0.0.0.0 203.0.113.2

```

```

BR-Main(config)# end
BR-Main# write memory

```

IP BGP summary

```

Router#show ip bgp summary
BGP router identifier 192.168.30.10, local AS number 100
BGP table version is 1, main routing table version 6
0 network entries using 0 bytes of memory
0 path entries using 0 bytes of memory
0/0 BGP path/bestpath attribute entries using 0 bytes of memory
0 BGP AS-PATH entries using 0 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 32 total bytes of memory
BGP activity 0/0 prefixes, 0/0 paths, scan interval 60 secs

Neighbor      V    AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
203.0.113.2    4 65000      0      0       1    0    0 07:17:36      4

```

IP Route

```

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 105.12.32.1 to network 0.0.0.0

10.0.0.0/30 is subnetted, 2 subnets
O   10.0.0.12/30 [110/128] via 192.168.30.9, 00:28:23, Serial0/3/1
O   10.0.0.15/30 [110/129] via 192.168.30.9, 00:28:23, Serial0/3/1
105.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C   105.12.32.0/29 is directly connected, GigabitEthernet0/1
L   105.12.32.2/32 is directly connected, GigabitEthernet0/1
192.168.1.0/27 is subnetted, 4 subnets
O   192.168.1.0/27 [110/3] via 192.168.20.1, 00:37:10, GigabitEthernet0/0
O   192.168.1.32/27 [110/3] via 192.168.20.1, 00:37:10, GigabitEthernet0/0
O   192.168.1.64/27 [110/3] via 192.168.20.1, 00:37:10, GigabitEthernet0/0
O   192.168.1.96/27 [110/3] via 192.168.20.1, 00:37:10, GigabitEthernet0/0
192.168.2.0/27 is subnetted, 4 subnets
O   192.168.2.0/27 [110/67] via 192.168.30.2, 00:05:01, Serial0/3/0
O   192.168.2.32/27 [110/67] via 192.168.30.2, 00:05:01, Serial0/3/0
O   192.168.2.64/27 [110/67] via 192.168.30.2, 00:05:01, Serial0/3/0
O   192.168.2.96/27 [110/67] via 192.168.30.2, 00:05:01, Serial0/3/0
192.168.3.0/27 is subnetted, 4 subnets
O   192.168.3.0/27 [110/67] via 192.168.30.9, 00:06:59, Serial0/3/1
O   192.168.3.32/27 [110/67] via 192.168.30.9, 00:06:59, Serial0/3/1
O   192.168.3.64/27 [110/67] via 192.168.30.9, 00:06:59, Serial0/3/1
O   192.168.3.96/27 [110/67] via 192.168.30.9, 00:06:59, Serial0/3/1
192.168.10.0/30 is subnetted, 3 subnets
O   192.168.10.0/30 [110/2] via 192.168.20.1, 00:37:10, GigabitEthernet0/0
O   192.168.10.4/30 [110/66] via 192.168.30.2, 00:05:01, Serial0/3/0
O   192.168.10.8/30 [110/66] via 192.168.30.9, 00:28:23, Serial0/3/1
192.168.20.0/24 is variably subnetted, 4 subnets, 2 masks
C   192.168.20.0/30 is directly connected, GigabitEthernet0/0
L   192.168.20.2/32 is directly connected, GigabitEthernet0/0
O   192.168.20.4/30 [110/65] via 192.168.30.2, 00:28:23, Serial0/3/0
O   192.168.20.8/30 [110/65] via 192.168.30.9, 00:28:23, Serial0/3/1
192.168.30.0/24 is variably subnetted, 5 subnets, 2 masks
C   192.168.30.0/30 is directly connected, Serial0/3/0
L   192.168.30.1/32 is directly connected, Serial0/3/0
O   192.168.30.4/30 [110/128] via 192.168.30.9, 00:28:23, Serial0/3/1
O   192.168.30.8/30 [110/128] via 192.168.30.2, 00:28:23, Serial0/3/0
C   192.168.30.8/30 is directly connected, Serial0/3/1

```

IP ospf neighbor

```

Router#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
11.11.11.11      1     FULL/DR         00:00:36    192.168.20.1   GigabitEthernet0/0
2.2.2.2          0     FULL/ -         00:00:35    192.168.30.2   Serial0/3/0
3.3.3.3          0     FULL/ -         00:00:36    192.168.30.9   Serial0/3/1
Router#

```

Router 3: BR-SC1 (Sub-Campus 1 Border Router – AS 65001)

Connects Sub-Campus 1 to ISP and forms the WAN triangle with BR-Main and BR-SC2.

```

Router> enable
Router# configure terminal

```

```

Router(config)# hostname BR-SC1

! Interface to ISP-Router
BR-SC1(config)# interface GigabitEthernet0/0
BR-SC1(config-if)# ip address 203.0.113.5 255.255.255.252
BR-SC1(config-if)# description Link-to-ISP
BR-SC1(config-if)# no shutdown
BR-SC1(config-if)# exit

! Interface to BR-Main (WAN Triangle)
BR-SC1(config)# interface GigabitEthernet0/1
BR-SC1(config-if)# ip address 10.0.0.2 255.255.255.252
BR-SC1(config-if)# description WAN-Link-to-BR-Main
BR-SC1(config-if)# no shutdown
BR-SC1(config-if)# exit

! Interface to BR-SC2 (WAN Triangle)
BR-SC1(config)# interface GigabitEthernet0/2
BR-SC1(config-if)# ip address 10.0.0.9 255.255.255.252
BR-SC1(config-if)# description WAN-Link-to-BR-SC2
BR-SC1(config-if)# no shutdown
BR-SC1(config-if)# exit

! Interface to IR-SC1 (Internal)
BR-SC1(config)# interface GigabitEthernet1/0
BR-SC1(config-if)# ip address 10.2.0.6 255.255.255.252
BR-SC1(config-if)# description Link-to-IR-SC1
BR-SC1(config-if)# no shutdown
BR-SC1(config-if)# exit

! BGP Configuration -- AS 65001
BR-SC1(config)# router bgp 65001
BR-SC1(config-router)# bgp router-id 2.2.2.2
BR-SC1(config-router)# neighbor 203.0.113.6 remote-as 65000
BR-SC1(config-router)# neighbor 10.0.0.1 remote-as 65000
BR-SC1(config-router)# neighbor 10.0.0.10 remote-as 65002
BR-SC1(config-router)# network 192.168.2.0 mask 255.255.255.224
BR-SC1(config-router)# network 192.168.2.32 mask 255.255.255.224
BR-SC1(config-router)# network 192.168.2.64 mask 255.255.255.224
BR-SC1(config-router)# network 192.168.2.96 mask 255.255.255.224
BR-SC1(config-router)# exit

! OSPF -- Internal area
BR-SC1(config)# router ospf 1
BR-SC1(config-router)# router-id 2.2.2.2
BR-SC1(config-router)# network 10.2.0.4 0.0.0.3 area 1
BR-SC1(config-router)# network 10.0.0.0 0.0.0.3 area 0
BR-SC1(config-router)# network 10.0.0.8 0.0.0.3 area 0
BR-SC1(config-router)# default-information originate

```

```
BR-SC1(config-router)# exit
```

```
! Default route to ISP
```

```
BR-SC1(config)# ip route 0.0.0.0 0.0.0.0 203.0.113.6
```

```
! ACL -- Block Admin from Web Server
```

```
BR-SC1(config)# ip access-list extended BLOCK_ADMIN_WEB
```

```
BR-SC1(config-ext-nacl)# deny tcp 192.168.2.64 0.0.0.31 host 192.168.1.5 eq 80
```

```
BR-SC1(config-ext-nacl)# deny tcp 192.168.2.64 0.0.0.31 host 192.168.1.5 eq 443
```

```
BR-SC1(config-ext-nacl)# permit ip any any
```

```
BR-SC1(config-ext-nacl)# exit
```

```
! Firewall -- Block unauthorized external inbound
```

```
BR-SC1(config)# ip access-list extended EXTERNAL_FIREWALL
```

```
BR-SC1(config-ext-nacl)# permit tcp any any established
```

```
BR-SC1(config-ext-nacl)# permit icmp any any
```

```
BR-SC1(config-ext-nacl)# deny ip any 192.168.2.0 0.0.0.255
```

```
BR-SC1(config-ext-nacl)# permit ip any any
```

```
BR-SC1(config-ext-nacl)# exit
```

```
BR-SC1(config)# interface GigabitEthernet0/0
```

```
BR-SC1(config-if)# ip access-group EXTERNAL_FIREWALL in
```

```
BR-SC1(config-if)# exit
```

```
BR-SC1(config)# end
```

```
BR-SC1# write memory
```

IP BGP summary

```
Router#show ip bgp summary
BGP router identifier 192.168.30.5, local AS number 200
BGP table version is 1, main routing table version 6
0 network entries using 0 bytes of memory
0 path entries using 0 bytes of memory
0/0 BGP path/bestpath attribute entries using 0 bytes of memory
0 BGP AS-PATH entries using 0 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 32 total bytes of memory
BGP activity 0/0 prefixes, 0/0 paths, scan interval 60 secs
```

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
203.0.113.6	4	65000	0	0	1	0	0	07:24:12	4

IP Route

```

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 105.12.32.9 to network 0.0.0.0

 10.0.0.0/30 is subnetted, 2 subnets
O   10.0.0.12/30 [110/128] via 192.168.30.6, 00:43:08, Serial0/3/1
O   10.10.15.0/30 [110/129] via 192.168.30.6, 00:34:41, Serial0/3/1
 105.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C   105.12.32.8/29 is directly connected, GigabitEthernet0/1
L   105.12.32.10/32 is directly connected, GigabitEthernet0/1
 192.168.1.0/27 is subnetted, 4 subnets
O   192.168.1.0/27 [110/67] via 192.168.30.1, 01:06:34, Serial0/3/0
O   192.168.1.32/27 [110/67] via 192.168.30.1, 01:06:34, Serial0/3/0
O   192.168.1.64/27 [110/67] via 192.168.30.1, 01:06:34, Serial0/3/0
O   192.168.1.96/27 [110/67] via 192.168.30.1, 01:06:34, Serial0/3/0
 192.168.2.0/27 is subnetted, 4 subnets
O   192.168.2.0/27 [110/3] via 192.168.20.5, 01:06:34, GigabitEthernet0/0
O   192.168.2.32/27 [110/3] via 192.168.20.5, 01:06:34, GigabitEthernet0/0
O   192.168.2.64/27 [110/3] via 192.168.20.5, 01:06:34, GigabitEthernet0/0
O   192.168.2.96/27 [110/3] via 192.168.20.5, 01:06:34, GigabitEthernet0/0
 192.168.3.0/27 is subnetted, 4 subnets
O   192.168.3.0/27 [110/67] via 192.168.30.6, 00:43:08, Serial0/3/1
O   192.168.3.32/27 [110/67] via 192.168.30.6, 00:43:08, Serial0/3/1
O   192.168.3.64/27 [110/67] via 192.168.30.6, 00:43:08, Serial0/3/1
O   192.168.3.96/27 [110/67] via 192.168.30.6, 00:43:08, Serial0/3/1
 192.168.10.0/30 is subnetted, 3 subnets
O   192.168.10.0/30 [110/66] via 192.168.30.1, 01:06:34, Serial0/3/0
O   192.168.10.4/30 [110/2] via 192.168.20.5, 01:06:34, GigabitEthernet0/0
O   192.168.10.8/30 [110/66] via 192.168.30.6, 00:43:08, Serial0/3/1
 192.168.20.0/24 is variably subnetted, 4 subnets, 2 masks
O   192.168.20.0/30 [110/65] via 192.168.30.1, 01:06:34, Serial0/3/0
C   192.168.20.4/30 is directly connected, GigabitEthernet0/0
L   192.168.20.6/32 is directly connected, GigabitEthernet0/0
O   192.168.20.8/30 [110/65] via 192.168.30.6, 00:43:08, Serial0/3/1
 192.168.30.0/24 is variably subnetted, 5 subnets, 2 masks
C   192.168.30.0/30 is directly connected, Serial0/3/0
L   192.168.30.2/32 is directly connected, Serial0/3/0
C   192.168.30.4/30 is directly connected, Serial0/3/1
L   192.168.30.5/32 is directly connected, Serial0/3/1
O   192.168.30.8/30 [110/128] via 192.168.30.1, 00:43:08, Serial0/3/0
    [110/128] via 192.168.30.6, 00:43:08, Serial0/3/1

```

Access List

```

Router#show access-lists
Standard IP access list 1
 10 permit 192.168.0.0 0.0.255.255 (2 match(es))

```

Router 4: BR-SC2 (Sub-Campus 2 Border Router – AS 65002)

Connects Sub-Campus 2 to ISP and completes the WAN triangle.

```

Router> enable
Router# configure terminal
Router(config)# hostname BR-SC2

! Interface to ISP-Router
BR-SC2(config)# interface GigabitEthernet0/0
BR-SC2(config-if)# ip address 203.0.113.9 255.255.255.252
BR-SC2(config-if)# description Link-to-ISP
BR-SC2(config-if)# no shutdown
BR-SC2(config-if)# exit

! Interface to BR-Main (WAN Triangle)
BR-SC2(config)# interface GigabitEthernet0/1
BR-SC2(config-if)# ip address 10.0.0.6 255.255.255.252
BR-SC2(config-if)# description WAN-Link-to-BR-Main
BR-SC2(config-if)# no shutdown
BR-SC2(config-if)# exit

! Interface to BR-SC1 (WAN Triangle)
BR-SC2(config)# interface GigabitEthernet0/2
BR-SC2(config-if)# ip address 10.0.0.10 255.255.255.252
BR-SC2(config-if)# description WAN-Link-to-BR-SC1
BR-SC2(config-if)# no shutdown
BR-SC2(config-if)# exit

! Interface to IR-SC2 (Internal)
BR-SC2(config)# interface GigabitEthernet1/0
BR-SC2(config-if)# ip address 10.3.0.6 255.255.255.252
BR-SC2(config-if)# description Link-to-IR-SC2
BR-SC2(config-if)# no shutdown
BR-SC2(config-if)# exit

! BGP Configuration -- AS 65002
BR-SC2(config)# router bgp 65002
BR-SC2(config-router)# bgp router-id 3.3.3.3
BR-SC2(config-router)# neighbor 203.0.113.10 remote-as 65000
BR-SC2(config-router)# neighbor 10.0.0.5 remote-as 65000
BR-SC2(config-router)# neighbor 10.0.0.9 remote-as 65001
BR-SC2(config-router)# network 192.168.3.0 mask 255.255.255.224
BR-SC2(config-router)# network 192.168.3.32 mask 255.255.255.224
BR-SC2(config-router)# network 192.168.3.64 mask 255.255.255.224
BR-SC2(config-router)# network 192.168.3.96 mask 255.255.255.224
BR-SC2(config-router)# exit

! OSPF -- Internal area
BR-SC2(config)# router ospf 1
BR-SC2(config-router)# router-id 3.3.3.3
BR-SC2(config-router)# network 10.3.0.4 0.0.0.3 area 2
BR-SC2(config-router)# network 10.0.0.4 0.0.0.3 area 0

```

```
BR-SC2(config-router)# network 10.0.0.8 0.0.0.3 area 0
BR-SC2(config-router)# default-information originate
BR-SC2(config-router)# exit
```

! Default route to ISP

```
BR-SC2(config)# ip route 0.0.0.0 0.0.0.0 203.0.113.10
```

! ACL -- Block Admin from Web Server

```
BR-SC2(config)# ip access-list extended BLOCK_ADMIN_WEB
```

```
BR-SC2(config-ext-nacl)# deny tcp 192.168.3.64 0.0.0.31 host 192.168.1.5 eq 80
```

```
BR-SC2(config-ext-nacl)# deny tcp 192.168.3.64 0.0.0.31 host 192.168.1.5 eq 443
```

```
BR-SC2(config-ext-nacl)# permit ip any any
```

```
BR-SC2(config-ext-nacl)# exit
```

! Firewall -- Block unauthorized external inbound

```
BR-SC2(config)# ip access-list extended EXTERNAL_FIREWALL
```

```
BR-SC2(config-ext-nacl)# permit tcp any any established
```

```
BR-SC2(config-ext-nacl)# permit icmp any any
```

```
BR-SC2(config-ext-nacl)# deny ip any 192.168.3.0 0.0.0.255
```

```
BR-SC2(config-ext-nacl)# permit ip any any
```

```
BR-SC2(config-ext-nacl)# exit
```

```
BR-SC2(config)# interface GigabitEthernet0/0
```

```
BR-SC2(config-if)# ip access-group EXTERNAL_FIREWALL in
```

```
BR-SC2(config-if)# exit
```

```
BR-SC2(config)# end
```

```
BR-SC2# write memory
```

IP BGP summary

```
Router#show ip bgp summary
BGP router identifier 192.168.30.9, local AS number 300
BGP table version is 1, main routing table version 6
0 network entries using 0 bytes of memory
0 path entries using 0 bytes of memory
0/0 BGP path/bestpath attribute entries using 0 bytes of memory
0 BGP AS-PATH entries using 0 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 32 total bytes of memory
BGP activity 0/0 prefixes, 0/0 paths, scan interval 60 secs
```

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
203.0.113.10	4	65000	0	0	1	0	0	01:38:13	4

IP Route

```

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 105.12.32.17 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       10.0.0.12/30 is directly connected, Serial0/2/0
L       10.0.0.14/32 is directly connected, Serial0/2/0
O       10.10.15.0/30 [110/65] via 10.0.0.13, 00:37:11, Serial0/2/0
    105.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       105.12.32.16/29 is directly connected, GigabitEthernet0/1
L       105.12.32.18/32 is directly connected, GigabitEthernet0/1
    192.168.1.0/27 is subnetted, 4 subnets
O       192.168.1.0/27 [110/67] via 192.168.30.10, 00:37:11, Serial0/3/1
O       192.168.1.32/27 [110/67] via 192.168.30.10, 00:37:11, Serial0/3/1
O       192.168.1.64/27 [110/67] via 192.168.30.10, 00:37:11, Serial0/3/1
O       192.168.1.96/27 [110/67] via 192.168.30.10, 00:37:11, Serial0/3/1
    192.168.2.0/27 is subnetted, 4 subnets
O       192.168.2.0/27 [110/67] via 192.168.30.5, 00:37:11, Serial0/3/0
O       192.168.2.32/27 [110/67] via 192.168.30.5, 00:37:11, Serial0/3/0
O       192.168.2.64/27 [110/67] via 192.168.30.5, 00:37:11, Serial0/3/0
O       192.168.2.96/27 [110/67] via 192.168.30.5, 00:37:11, Serial0/3/0
    192.168.3.0/27 is subnetted, 4 subnets
O       192.168.3.0/27 [110/3] via 192.168.20.9, 00:37:11, GigabitEthernet0/0
O       192.168.3.32/27 [110/3] via 192.168.20.9, 00:37:11, GigabitEthernet0/0
O       192.168.3.64/27 [110/3] via 192.168.20.9, 00:37:11, GigabitEthernet0/0
O       192.168.3.96/27 [110/3] via 192.168.20.9, 00:37:11, GigabitEthernet0/0
    192.168.10.0/30 is subnetted, 3 subnets
O       192.168.10.0/30 [110/66] via 192.168.30.10, 00:37:11, Serial0/3/1
O       192.168.10.4/30 [110/66] via 192.168.30.5, 00:37:11, Serial0/3/0
O       192.168.10.8/30 [110/2] via 192.168.20.9, 00:37:11, GigabitEthernet0/0
    192.168.20.0/24 is variably subnetted, 4 subnets, 2 masks
O       192.168.20.0/30 [110/65] via 192.168.30.10, 00:37:11, Serial0/3/1
O       192.168.20.4/30 [110/65] via 192.168.30.5, 00:37:11, Serial0/3/0
C       192.168.20.8/30 is directly connected, GigabitEthernet0/0
L       192.168.20.10/32 is directly connected, GigabitEthernet0/0
    192.168.30.0/24 is variably subnetted, 5 subnets, 2 masks
O       192.168.30.0/30 [110/128] via 192.168.30.10, 00:37:11, Serial0/3/1
O       192.168.30.4/30 [110/128] via 192.168.30.5, 00:37:11, Serial0/3/0
C       192.168.30.4/30 is directly connected, Serial0/3/0
L       192.168.30.6/32 is directly connected, Serial0/3/0
C       192.168.30.8/30 is directly connected, Serial0/3/1

```

Router 5: IR-Main (Internal Router – Main Campus)

Sits between the L3 Core Switch and BR-Main. Handles OSPF redistribution for the main campus subnets.

```

Router> enable
Router# configure terminal
Router(config)# hostname IR-Main

```

```

! Interface to L3-Core-SW
IR-Main(config)# interface GigabitEthernet0/0
IR-Main(config-if)# ip address 10.1.0.1 255.255.255.252
IR-Main(config-if)# description Link-to-L3-Core-SW
IR-Main(config-if)# no shutdown
IR-Main(config-if)# exit

```

```

! Interface to BR-Main

```

```
IR-Main(config)# interface GigabitEthernet0/1
IR-Main(config-if)# ip address 10.1.0.5 255.255.255.252
IR-Main(config-if)# description Link-to-BR-Main
IR-Main(config-if)# no shutdown
IR-Main(config-if)# exit
```

! OSPF Configuration

```
IR-Main(config)# router ospf 1
IR-Main(config-router)# router-id 5.5.5.5
IR-Main(config-router)# network 10.1.0.0 0.0.0.3 area 0
IR-Main(config-router)# network 10.1.0.4 0.0.0.3 area 0
IR-Main(config-router)# network 192.168.1.0 0.0.0.31 area 0
IR-Main(config-router)# network 192.168.1.32 0.0.0.31 area 0
IR-Main(config-router)# network 192.168.1.64 0.0.0.31 area 0
IR-Main(config-router)# network 192.168.1.96 0.0.0.31 area 0
IR-Main(config-router)# exit
```

```
IR-Main(config)# end
IR-Main# write memory
```

IP OSPF Neighbor

Router#show ip ospf neighbor					
Neighbor ID	Pri	State	Dead Time	Address	Interface
1.1.1.1	1	FULL/BDR	00:00:32	192.168.20.2	GigabitEthernet0/0/1
192.168.10.1	1	FULL/DR	00:00:32	192.168.10.1	GigabitEthernet0/0/0

IP Route

```

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 192.168.20.2 to network 0.0.0.0

 10.0.0.0/30 is subnetted, 2 subnets
O   10.0.0.12/30 [110/129] via 192.168.20.2, 00:46:01, GigabitEthernet0/0/1
O   10.10.15.0/30 [110/130] via 192.168.20.2, 00:46:01, GigabitEthernet0/0/1
 192.168.1.0/27 is subnetted, 4 subnets
O   192.168.1.0/27 [110/2] via 192.168.10.1, 01:17:54, GigabitEthernet0/0/0
O   192.168.1.32/27 [110/2] via 192.168.10.1, 01:17:54, GigabitEthernet0/0/0
O   192.168.1.64/27 [110/2] via 192.168.10.1, 01:17:54, GigabitEthernet0/0/0
O   192.168.1.96/27 [110/2] via 192.168.10.1, 01:17:54, GigabitEthernet0/0/0
 192.168.2.0/27 is subnetted, 4 subnets
O   192.168.2.0/27 [110/68] via 192.168.20.2, 00:24:37, GigabitEthernet0/0/1
O   192.168.2.32/27 [110/68] via 192.168.20.2, 00:24:37, GigabitEthernet0/0/1
O   192.168.2.64/27 [110/68] via 192.168.20.2, 00:24:37, GigabitEthernet0/0/1
O   192.168.2.96/27 [110/68] via 192.168.20.2, 00:24:37, GigabitEthernet0/0/1
 192.168.3.0/27 is subnetted, 4 subnets
O   192.168.3.0/27 [110/68] via 192.168.20.2, 00:24:37, GigabitEthernet0/0/1
O   192.168.3.32/27 [110/68] via 192.168.20.2, 00:24:37, GigabitEthernet0/0/1
O   192.168.3.64/27 [110/68] via 192.168.20.2, 00:24:37, GigabitEthernet0/0/1
O   192.168.3.96/27 [110/68] via 192.168.20.2, 00:24:37, GigabitEthernet0/0/1
 192.168.10.0/24 is variably subnetted, 4 subnets, 2 masks
C   192.168.10.0/30 is directly connected, GigabitEthernet0/0/0
L   192.168.10.2/32 is directly connected, GigabitEthernet0/0/0
O   192.168.10.4/30 [110/67] via 192.168.20.2, 00:45:51, GigabitEthernet0/0/1
O   192.168.10.8/30 [110/67] via 192.168.20.2, 00:46:01, GigabitEthernet0/0/1
 192.168.20.0/24 is variably subnetted, 4 subnets, 2 masks
C   192.168.20.0/30 is directly connected, GigabitEthernet0/0/1
L   192.168.20.1/32 is directly connected, GigabitEthernet0/0/1
O   192.168.20.4/30 [110/66] via 192.168.20.2, 00:45:51, GigabitEthernet0/0/1
O   192.168.20.8/30 [110/66] via 192.168.20.2, 00:46:01, GigabitEthernet0/0/1
 192.168.30.0/30 is subnetted, 3 subnets
O   192.168.30.0/30 [110/65] via 192.168.20.2, 00:54:33, GigabitEthernet0/0/1
O   192.168.30.4/30 [110/129] via 192.168.20.2, 00:46:01, GigabitEthernet0/0/1
O   192.168.30.8/30 [110/65] via 192.168.20.2, 00:54:33, GigabitEthernet0/0/1
O*E2 0.0.0.0/0 [110/1] via 192.168.20.2, 01:17:54, GigabitEthernet0/0/1

Router#

```

Router 6: IR-SC1 (Internal Router – Sub-Campus 1)

Sits between L3-SW-SC1 and BR-SC1. Routes internal OSPF traffic for Sub-Campus 1 subnets.

```

Router> enable
Router# configure terminal
Router(config)# hostname IR-SC1

```

```

! Interface to L3-SW-SC1
IR-SC1(config)# interface GigabitEthernet0/0
IR-SC1(config-if)# ip address 10.2.0.1 255.255.255.252

```

```
IR-SC1(config-if)# description Link-to-L3-SW-SC1
IR-SC1(config-if)# no shutdown
IR-SC1(config-if)# exit
```

! Interface to BR-SC1

```
IR-SC1(config)# interface GigabitEthernet0/1
IR-SC1(config-if)# ip address 10.2.0.5 255.255.255.252
IR-SC1(config-if)# description Link-to-BR-SC1
IR-SC1(config-if)# no shutdown
IR-SC1(config-if)# exit
```

! OSPF Configuration

```
IR-SC1(config)# router ospf 1
IR-SC1(config-router)# router-id 6.6.6.6
IR-SC1(config-router)# network 10.2.0.0 0.0.0.3 area 1
IR-SC1(config-router)# network 10.2.0.4 0.0.0.3 area 1
IR-SC1(config-router)# network 192.168.2.0 0.0.0.31 area 1
IR-SC1(config-router)# network 192.168.2.32 0.0.0.31 area 1
IR-SC1(config-router)# network 192.168.2.64 0.0.0.31 area 1
IR-SC1(config-router)# network 192.168.2.96 0.0.0.31 area 1
IR-SC1(config-router)# exit
```

```
IR-SC1(config)# end
IR-SC1# write memory
```

IP OSPF Neighbor

```
Router#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.10.5	1	FULL/DR	00:00:36	192.168.10.5	GigabitEthernet0/0/0
2.2.2.2	1	FULL/BDR	00:00:37	192.168.20.6	GigabitEthernet0/0/1

IP Route

```

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 192.168.20.6 to network 0.0.0.0

 10.0.0.0/30 is subnetted, 2 subnets
O   10.0.0.12/30 [110/129] via 192.168.20.6, 00:48:14, GigabitEthernet0/0/1
O   10.10.15.0/30 [110/130] via 192.168.20.6, 00:48:14, GigabitEthernet0/0/1
 192.168.1.0/27 is subnetted, 4 subnets
O   192.168.1.0/27 [110/68] via 192.168.20.6, 00:48:14, GigabitEthernet0/0/1
O   192.168.1.32/27 [110/68] via 192.168.20.6, 00:48:14, GigabitEthernet0/0/1
O   192.168.1.64/27 [110/68] via 192.168.20.6, 00:48:14, GigabitEthernet0/0/1
O   192.168.1.96/27 [110/68] via 192.168.20.6, 00:48:14, GigabitEthernet0/0/1
 192.168.2.0/27 is subnetted, 4 subnets
O   192.168.2.0/27 [110/2] via 192.168.10.5, 07:37:00, GigabitEthernet0/0/0
O   192.168.2.32/27 [110/2] via 192.168.10.5, 07:37:00, GigabitEthernet0/0/0
O   192.168.2.64/27 [110/2] via 192.168.10.5, 07:37:00, GigabitEthernet0/0/0
O   192.168.2.96/27 [110/2] via 192.168.10.5, 07:37:00, GigabitEthernet0/0/0
 192.168.3.0/27 is subnetted, 4 subnets
O   192.168.3.0/27 [110/68] via 192.168.20.6, 00:48:14, GigabitEthernet0/0/1
O   192.168.3.32/27 [110/68] via 192.168.20.6, 00:48:14, GigabitEthernet0/0/1
O   192.168.3.64/27 [110/68] via 192.168.20.6, 00:48:14, GigabitEthernet0/0/1
O   192.168.3.96/27 [110/68] via 192.168.20.6, 00:48:14, GigabitEthernet0/0/1
 192.168.10.0/24 is variably subnetted, 4 subnets, 2 masks
O   192.168.10.0/30 [110/67] via 192.168.20.6, 00:48:14, GigabitEthernet0/0/1
C   192.168.10.4/30 is directly connected, GigabitEthernet0/0/0
L   192.168.10.6/32 is directly connected, GigabitEthernet0/0/0
O   192.168.10.8/30 [110/67] via 192.168.20.6, 00:48:14, GigabitEthernet0/0/1
 192.168.20.0/24 is variably subnetted, 4 subnets, 2 masks
O   192.168.20.0/30 [110/66] via 192.168.20.6, 00:48:14, GigabitEthernet0/0/1
C   192.168.20.4/30 is directly connected, GigabitEthernet0/0/1
L   192.168.20.5/32 is directly connected, GigabitEthernet0/0/1
O   192.168.20.8/30 [110/66] via 192.168.20.6, 00:48:14, GigabitEthernet0/0/1
 192.168.30.0/30 is subnetted, 3 subnets
O   192.168.30.0/30 [110/65] via 192.168.20.6, 01:59:31, GigabitEthernet0/0/1
O   192.168.30.4/30 [110/65] via 192.168.20.6, 01:49:36, GigabitEthernet0/0/1
O   192.168.30.8/30 [110/129] via 192.168.20.6, 00:48:14, GigabitEthernet0/0/1
O*E2 0.0.0.0/0 [110/1] via 192.168.20.6, 00:48:14, GigabitEthernet0/0/1

```

Router 7: IR-SC2 (Internal Router – Sub-Campus 2)

Sits between L3-SW-SC2 and BR-SC2. Routes internal OSPF traffic for Sub-Campus 2 subnets.

```

Router> enable
Router# configure terminal
Router(config)# hostname IR-SC2

```

! Interface to L3-SW-SC2

```
IR-SC2(config)# interface GigabitEthernet0/0
IR-SC2(config-if)# ip address 10.3.0.1 255.255.255.252
IR-SC2(config-if)# description Link-to-L3-SW-SC2
IR-SC2(config-if)# no shutdown
IR-SC2(config-if)# exit
```

! Interface to BR-SC2

```
IR-SC2(config)# interface GigabitEthernet0/1
IR-SC2(config-if)# ip address 10.3.0.5 255.255.255.252
IR-SC2(config-if)# description Link-to-BR-SC2
IR-SC2(config-if)# no shutdown
IR-SC2(config-if)# exit
```

! OSPF Configuration

```
IR-SC2(config)# router ospf 1
IR-SC2(config-router)# router-id 7.7.7.7
IR-SC2(config-router)# network 10.3.0.0 0.0.0.3 area 2
IR-SC2(config-router)# network 10.3.0.4 0.0.0.3 area 2
IR-SC2(config-router)# network 192.168.3.0 0.0.0.31 area 2
IR-SC2(config-router)# network 192.168.3.32 0.0.0.31 area 2
IR-SC2(config-router)# network 192.168.3.64 0.0.0.31 area 2
IR-SC2(config-router)# network 192.168.3.96 0.0.0.31 area 2
IR-SC2(config-router)# exit
```

```
IR-SC2(config)# end
IR-SC2# write memory
```

IP OSPF Neighbor

```
Router#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
33.33.33.33	1	FULL/DR	00:00:38	192.168.20.9	GigabitEthernet0/0
2.2.2.2	0	FULL/ -	00:00:38	192.168.30.5	Serial0/3/0
1.1.1.1	0	FULL/ -	00:00:37	192.168.30.10	Serial0/3/1
10.10.15.2	0	FULL/ -	00:00:37	10.0.0.13	Serial0/2/0

IP Route

```

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 105.12.32.17 to network 0.0.0.0

10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C    10.0.0.12/30 is directly connected, Serial0/2/0
L    10.0.0.14/32 is directly connected, Serial0/2/0
O    10.10.15.0/30 [110/65] via 10.0.0.13, 00:49:45, Serial0/2/0
105.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    105.12.32.16/29 is directly connected, GigabitEthernet0/1
L    105.12.32.18/32 is directly connected, GigabitEthernet0/1
192.168.1.0/27 is subnetted, 4 subnets
O    192.168.1.0/27 [110/67] via 192.168.30.10, 00:49:45, Serial0/3/1
O    192.168.1.32/27 [110/67] via 192.168.30.10, 00:49:45, Serial0/3/1
O    192.168.1.64/27 [110/67] via 192.168.30.10, 00:49:45, Serial0/3/1
O    192.168.1.96/27 [110/67] via 192.168.30.10, 00:49:45, Serial0/3/1
192.168.2.0/27 is subnetted, 4 subnets
O    192.168.2.0/27 [110/67] via 192.168.30.5, 00:49:45, Serial0/3/0
O    192.168.2.32/27 [110/67] via 192.168.30.5, 00:49:45, Serial0/3/0
O    192.168.2.64/27 [110/67] via 192.168.30.5, 00:49:45, Serial0/3/0
O    192.168.2.96/27 [110/67] via 192.168.30.5, 00:49:45, Serial0/3/0
192.168.3.0/27 is subnetted, 4 subnets
O    192.168.3.0/27 [110/3] via 192.168.20.9, 00:49:45, GigabitEthernet0/0
O    192.168.3.32/27 [110/3] via 192.168.20.9, 00:49:45, GigabitEthernet0/0
O    192.168.3.64/27 [110/3] via 192.168.20.9, 00:49:45, GigabitEthernet0/0
O    192.168.3.96/27 [110/3] via 192.168.20.9, 00:49:45, GigabitEthernet0/0
192.168.10.0/30 is subnetted, 3 subnets
O    192.168.10.0/30 [110/66] via 192.168.30.10, 00:49:45, Serial0/3/1
O    192.168.10.4/30 [110/66] via 192.168.30.5, 00:49:45, Serial0/3/0
O    192.168.10.8/30 [110/2] via 192.168.20.9, 00:49:45, GigabitEthernet0/0
192.168.20.0/24 is variably subnetted, 4 subnets, 2 masks
O    192.168.20.0/30 [110/65] via 192.168.30.10, 00:49:45, Serial0/3/1
O    192.168.20.4/30 [110/65] via 192.168.30.5, 00:49:45, Serial0/3/0
C    192.168.20.8/30 is directly connected, GigabitEthernet0/0
L    192.168.20.10/32 is directly connected, GigabitEthernet0/0
192.168.30.0/24 is variably subnetted, 5 subnets, 2 masks
O    192.168.30.0/30 [110/128] via 192.168.30.10, 00:49:45, Serial0/3/1
        [110/128] via 192.168.30.5, 00:49:45, Serial0/3/0
C    192.168.30.4/30 is directly connected, Serial0/3/0
L    192.168.30.6/32 is directly connected, Serial0/3/0
C    192.168.30.8/30 is directly connected, Serial0/3/1

```

Router 8: IR-Server (Server Farm Dedicated Router – Main Campus)

Dedicated router providing controlled access to the central server farm. Connects to L3-Core-SW and advertises server subnets via OSPF.

```

Router> enable
Router# configure terminal
Router(config)# hostname IR-Server

```

```

! Interface to L3-Core-SW (Server side)
IR-Server(config)# interface GigabitEthernet0/0
IR-Server(config-if)# ip address 10.1.0.9 255.255.255.252
IR-Server(config-if)# description Link-to-L3-Core-SW-ServerPort
IR-Server(config-if)# no shutdown
IR-Server(config-if)# exit

! Interface directly to Server Farm subnet
IR-Server(config)# interface GigabitEthernet0/1
IR-Server(config-if)# ip address 192.168.1.1 255.255.255.224
IR-Server(config-if)# description Server-Farm-Gateway
IR-Server(config-if)# no shutdown
IR-Server(config-if)# exit

! OSPF Configuration
IR-Server(config)# router ospf 1
IR-Server(config-router)# router-id 8.8.8.8
IR-Server(config-router)# network 10.1.0.8 0.0.0.3 area 0
IR-Server(config-router)# network 192.168.1.0 0.0.0.31 area 0
IR-Server(config-router)# exit

! ACL -- Block Admin from Web Server (Port 80/443)
IR-Server(config)# ip access-list extended PROTECT_SERVER_FARM
IR-Server(config-ext-nacl)# deny tcp 192.168.1.64 0.0.0.31 host 192.168.1.5 e
q 80
IR-Server(config-ext-nacl)# deny tcp 192.168.1.64 0.0.0.31 host 192.168.1.5 e
q 443
IR-Server(config-ext-nacl)# deny tcp 192.168.2.64 0.0.0.31 host 192.168.1.5 e
q 80
IR-Server(config-ext-nacl)# deny tcp 192.168.2.64 0.0.0.31 host 192.168.1.5 e
q 443
IR-Server(config-ext-nacl)# deny tcp 192.168.3.64 0.0.0.31 host 192.168.1.5 e
q 80
IR-Server(config-ext-nacl)# deny tcp 192.168.3.64 0.0.0.31 host 192.168.1.5 e
q 443
IR-Server(config-ext-nacl)# permit ip any any
IR-Server(config-ext-nacl)# exit

IR-Server(config)# interface GigabitEthernet0/1
IR-Server(config-if)# ip access-group PROTECT_SERVER_FARM in
IR-Server(config-if)# exit

IR-Server(config)# end
IR-Server# write memory

IR-Server# show ip ospf neighbor IR-Server# show ip route IR-Server# show
access-lists

```

8. Switch & L3 Switch Configuration

8.1 L3-Core-SW (Main Campus Layer 3 Switch)

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname L3-Core-SW
Switch(config)# ip routing

! Create VLANs
L3-Core-SW(config)# vlan 10
L3-Core-SW(config-vlan)# name CS_DEPT
L3-Core-SW(config-vlan)# exit
L3-Core-SW(config)# vlan 20
L3-Core-SW(config-vlan)# name EE_DEPT
L3-Core-SW(config-vlan)# exit
L3-Core-SW(config)# vlan 30
L3-Core-SW(config-vlan)# name ADMIN_DEPT
L3-Core-SW(config-vlan)# exit
L3-Core-SW(config)# vlan 40
L3-Core-SW(config-vlan)# name WIRELESS_MAIN
L3-Core-SW(config-vlan)# exit
L3-Core-SW(config)# vlan 50
L3-Core-SW(config-vlan)# name SERVER_FARM
L3-Core-SW(config-vlan)# exit

! SVI Interfaces
L3-Core-SW(config)# interface vlan 10
L3-Core-SW(config-if)# ip address 192.168.1.1 255.255.255.224
L3-Core-SW(config-if)# no shutdown
L3-Core-SW(config-if)# exit

L3-Core-SW(config)# interface vlan 20
L3-Core-SW(config-if)# ip address 192.168.1.33 255.255.255.224
L3-Core-SW(config-if)# no shutdown
L3-Core-SW(config-if)# exit

L3-Core-SW(config)# interface vlan 30
L3-Core-SW(config-if)# ip address 192.168.1.65 255.255.255.224
L3-Core-SW(config-if)# no shutdown
L3-Core-SW(config-if)# exit

L3-Core-SW(config)# interface vlan 40
L3-Core-SW(config-if)# ip address 192.168.1.97 255.255.255.224
L3-Core-SW(config-if)# no shutdown
L3-Core-SW(config-if)# exit

! Routed port to IR-Main
L3-Core-SW(config)# interface GigabitEthernet0/1
```

```

L3-Core-SW(config-if)# no switchport
L3-Core-SW(config-if)# ip address 10.1.0.2 255.255.255.252
L3-Core-SW(config-if)# no shutdown
L3-Core-SW(config-if)# exit

! Routed port to IR-Server
L3-Core-SW(config)# interface GigabitEthernet0/2
L3-Core-SW(config-if)# no switchport
L3-Core-SW(config-if)# ip address 10.1.0.10 255.255.255.252
L3-Core-SW(config-if)# no shutdown
L3-Core-SW(config-if)# exit

! Trunk port to SW-Main
L3-Core-SW(config)# interface GigabitEthernet0/3
L3-Core-SW(config-if)# switchport trunk encapsulation dot1q
L3-Core-SW(config-if)# switchport mode trunk
L3-Core-SW(config-if)# switchport trunk allowed vlan 10,20,30,40,50,99
L3-Core-SW(config-if)# exit

! OSPF
L3-Core-SW(config)# router ospf 1
L3-Core-SW(config-router)# router-id 4.4.4.4
L3-Core-SW(config-router)# network 192.168.1.0 0.0.0.31 area 0
L3-Core-SW(config-router)# network 192.168.1.32 0.0.0.31 area 0
L3-Core-SW(config-router)# network 192.168.1.64 0.0.0.31 area 0
L3-Core-SW(config-router)# network 192.168.1.96 0.0.0.31 area 0
L3-Core-SW(config-router)# network 10.1.0.0 0.0.0.3 area 0
L3-Core-SW(config-router)# network 10.1.0.8 0.0.0.3 area 0
L3-Core-SW(config-router)# exit

L3-Core-SW(config)# end
L3-Core-SW# write memory

```

8.2 L3-SW-SC1 (Sub-Campus 1 Layer 3 Switch)

```

Switch> enable
Switch# configure terminal
Switch(config)# hostname L3-SW-SC1
Switch(config)# ip routing

! Create VLANs
L3-SW-SC1(config)# vlan 10
L3-SW-SC1(config-vlan)# name CS_DEPT
L3-SW-SC1(config-vlan)# exit
L3-SW-SC1(config)# vlan 20
L3-SW-SC1(config-vlan)# name EE_DEPT
L3-SW-SC1(config-vlan)# exit
L3-SW-SC1(config)# vlan 30
L3-SW-SC1(config-vlan)# name ADMIN_DEPT
L3-SW-SC1(config-vlan)# exit

```

```

L3-SW-SC1(config)# vlan 41
L3-SW-SC1(config-vlan)# name WIRELESS_SC1
L3-SW-SC1(config-vlan)# exit

! SVI Interfaces
L3-SW-SC1(config)# interface vlan 10
L3-SW-SC1(config-if)# ip address 192.168.2.1 255.255.255.224
L3-SW-SC1(config-if)# no shutdown
L3-SW-SC1(config-if)# exit

L3-SW-SC1(config)# interface vlan 20
L3-SW-SC1(config-if)# ip address 192.168.2.33 255.255.255.224
L3-SW-SC1(config-if)# no shutdown
L3-SW-SC1(config-if)# exit

L3-SW-SC1(config)# interface vlan 30
L3-SW-SC1(config-if)# ip address 192.168.2.65 255.255.255.224
L3-SW-SC1(config-if)# no shutdown
L3-SW-SC1(config-if)# exit

L3-SW-SC1(config)# interface vlan 41
L3-SW-SC1(config-if)# ip address 192.168.2.97 255.255.255.224
L3-SW-SC1(config-if)# no shutdown
L3-SW-SC1(config-if)# exit

! Routed port to IR-SC1
L3-SW-SC1(config)# interface GigabitEthernet0/1
L3-SW-SC1(config-if)# no switchport
L3-SW-SC1(config-if)# ip address 10.2.0.2 255.255.255.252
L3-SW-SC1(config-if)# no shutdown
L3-SW-SC1(config-if)# exit

! Trunk to SW-SC1
L3-SW-SC1(config)# interface GigabitEthernet0/2
L3-SW-SC1(config-if)# switchport trunk encapsulation dot1q
L3-SW-SC1(config-if)# switchport mode trunk
L3-SW-SC1(config-if)# switchport trunk allowed vlan 10,20,30,41,99
L3-SW-SC1(config-if)# exit

! OSPF
L3-SW-SC1(config)# router ospf 1
L3-SW-SC1(config-router)# router-id 5.5.5.2
L3-SW-SC1(config-router)# network 192.168.2.0 0.0.0.31 area 1
L3-SW-SC1(config-router)# network 192.168.2.32 0.0.0.31 area 1
L3-SW-SC1(config-router)# network 192.168.2.64 0.0.0.31 area 1
L3-SW-SC1(config-router)# network 192.168.2.96 0.0.0.31 area 1
L3-SW-SC1(config-router)# network 10.2.0.0 0.0.0.3 area 1
L3-SW-SC1(config-router)# exit

```

```

! ACL on Admin VLAN 30 SVI
L3-SW-SC1(config)# ip access-list extended BLOCK_ADMIN_HTTP
L3-SW-SC1(config-ext-nacl)# deny tcp 192.168.2.64 0.0.0.31 host 192.168.1.5 eq 80
L3-SW-SC1(config-ext-nacl)# deny tcp 192.168.2.64 0.0.0.31 host 192.168.1.5 eq 443
L3-SW-SC1(config-ext-nacl)# permit ip any any
L3-SW-SC1(config-ext-nacl)# exit

L3-SW-SC1(config)# interface vlan 30
L3-SW-SC1(config-if)# ip access-group BLOCK_ADMIN_HTTP in
L3-SW-SC1(config-if)# exit

L3-SW-SC1(config)# end
L3-SW-SC1# write memory

```

8.3 Access Switch Configuration (SW-SC1)

```

Switch> enable
Switch# configure terminal
Switch(config)# hostname SW-SC1

SW-SC1(config)# vlan 10
SW-SC1(config-vlan)# name CS_DEPT
SW-SC1(config-vlan)# exit
SW-SC1(config)# vlan 20
SW-SC1(config-vlan)# name EE_DEPT
SW-SC1(config-vlan)# exit
SW-SC1(config)# vlan 30
SW-SC1(config-vlan)# name ADMIN_DEPT
SW-SC1(config-vlan)# exit
SW-SC1(config)# vlan 41
SW-SC1(config-vlan)# name WIRELESS_SC1
SW-SC1(config-vlan)# exit

! CS Ports
SW-SC1(config)# interface range FastEthernet0/1 - 10
SW-SC1(config-if-range)# switchport mode access
SW-SC1(config-if-range)# switchport access vlan 10
SW-SC1(config-if-range)# exit

! EE Ports
SW-SC1(config)# interface range FastEthernet0/11 - 20
SW-SC1(config-if-range)# switchport mode access
SW-SC1(config-if-range)# switchport access vlan 20
SW-SC1(config-if-range)# exit

! Admin Ports
SW-SC1(config)# interface range FastEthernet0/21 - 30

```

```

SW-SC1(config-if-range)# switchport mode access
SW-SC1(config-if-range)# switchport access vlan 30
SW-SC1(config-if-range)# exit

! Trunk to L3-SW-SC1
SW-SC1(config)# interface GigabitEthernet0/1
SW-SC1(config-if)# switchport mode trunk
SW-SC1(config-if)# switchport trunk allowed vlan 10,20,30,41,99
SW-SC1(config-if)# exit

SW-SC1(config)# end
SW-SC1# write memory

```

VLAN Brief

```

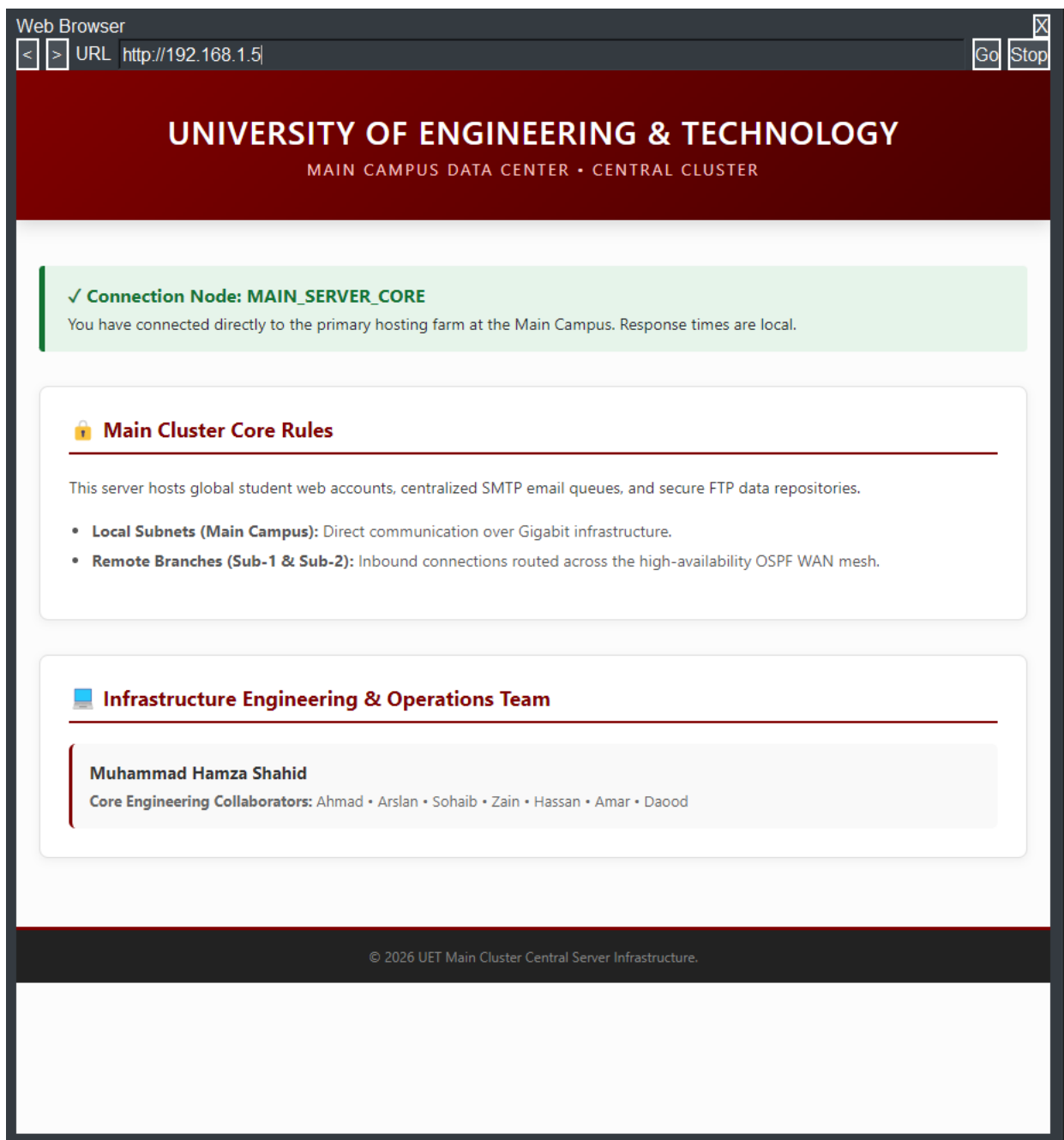
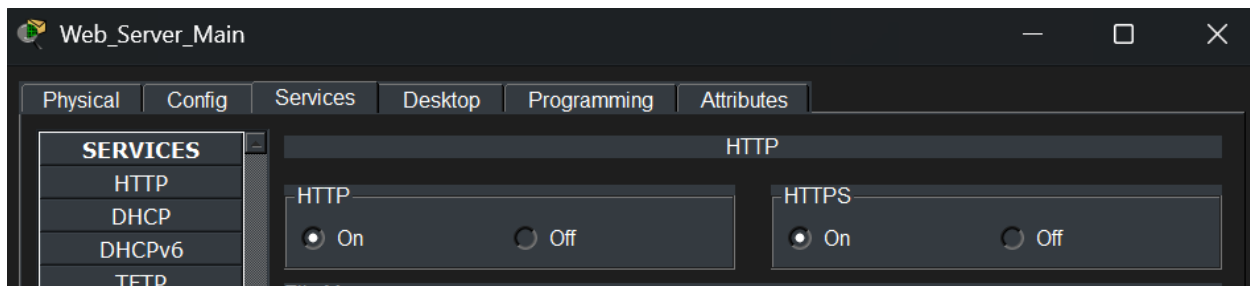
Switch#show vlan brief

```

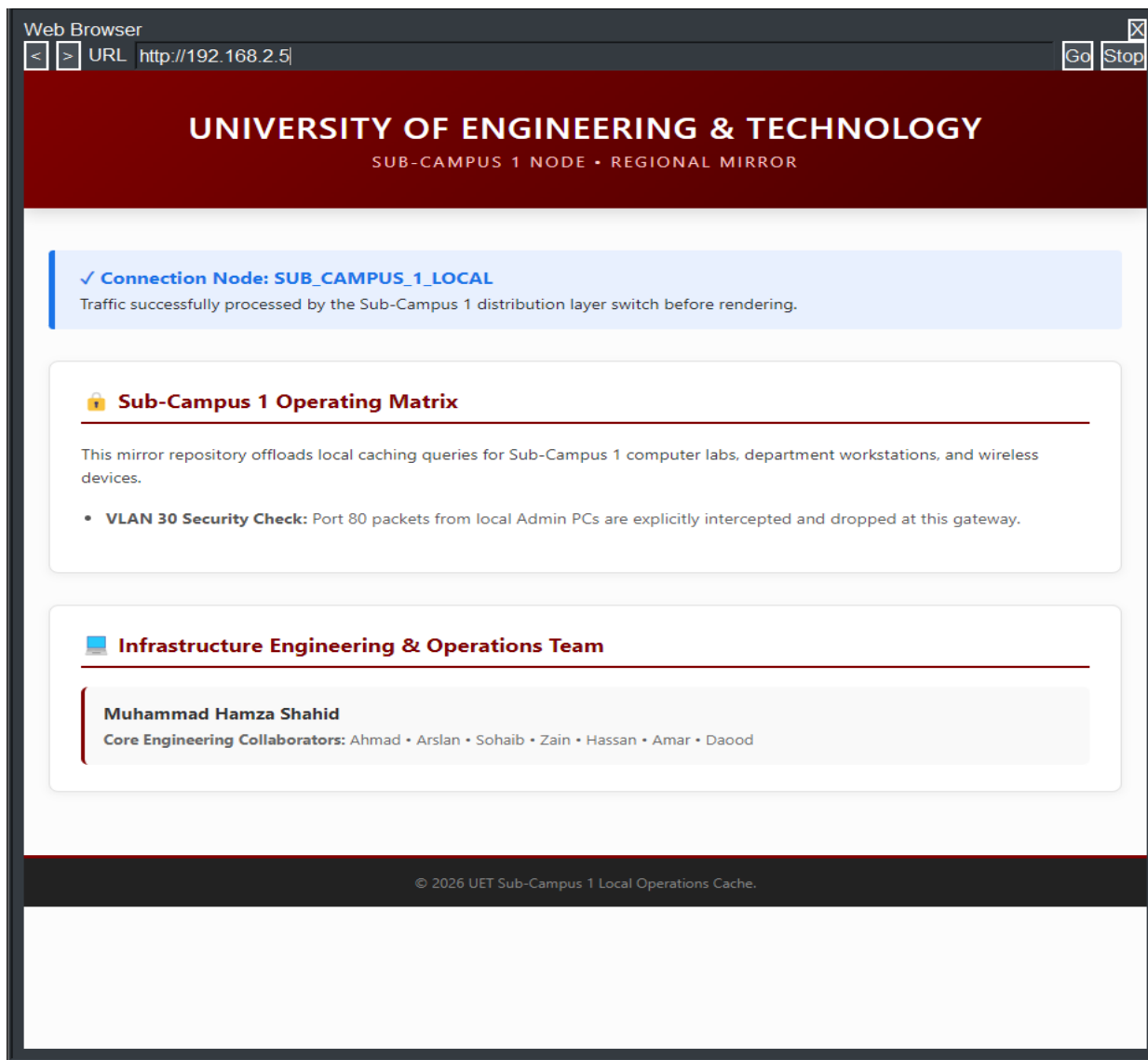
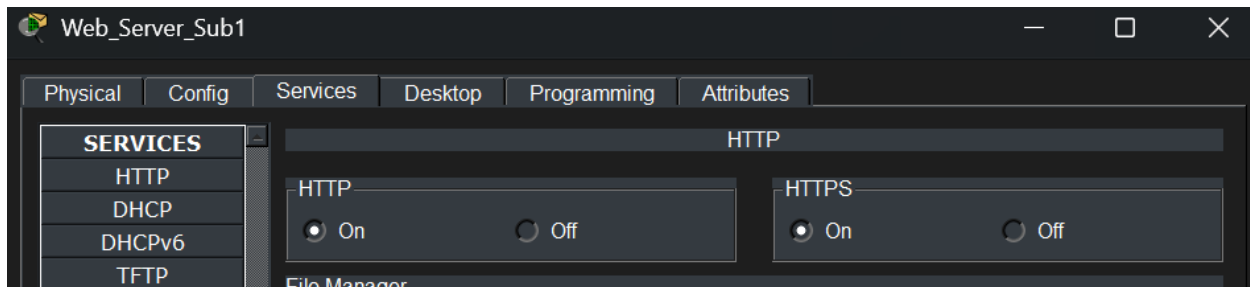
VLAN	Name	Status	Ports
1	default	active	Gig1/0/8, Gig1/0/9, Gig1/0/10, Gig1/0/11, Gig1/0/12, Gig1/0/13, Gig1/0/14, Gig1/0/15, Gig1/0/16, Gig1/0/17, Gig1/0/18, Gig1/0/19, Gig1/0/20, Gig1/0/21, Gig1/0/22, Gig1/0/23, Gig1/0/24, Gig1/1/1, Gig1/1/2, Gig1/1/3, Gig1/1/4
10	Computer_Science	active	Gig1/0/2, Gig1/0/6, Gig1/0/7
20	Electrical_Engineering	active	Gig1/0/3
30	Administration	active	Gig1/0/4
40	Wireless_Traffic	active	Gig1/0/5
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

9. Server Farm Configuration

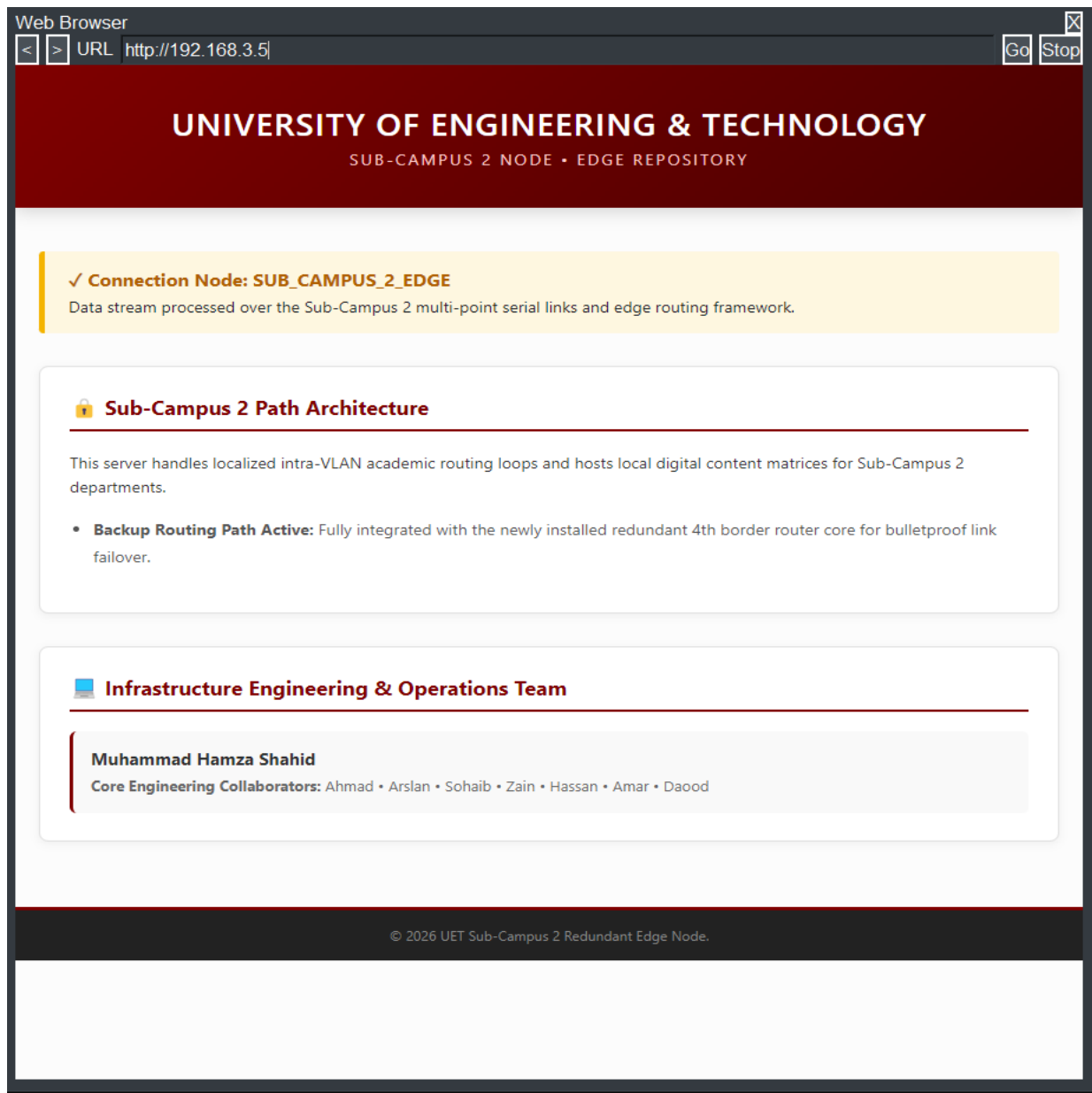
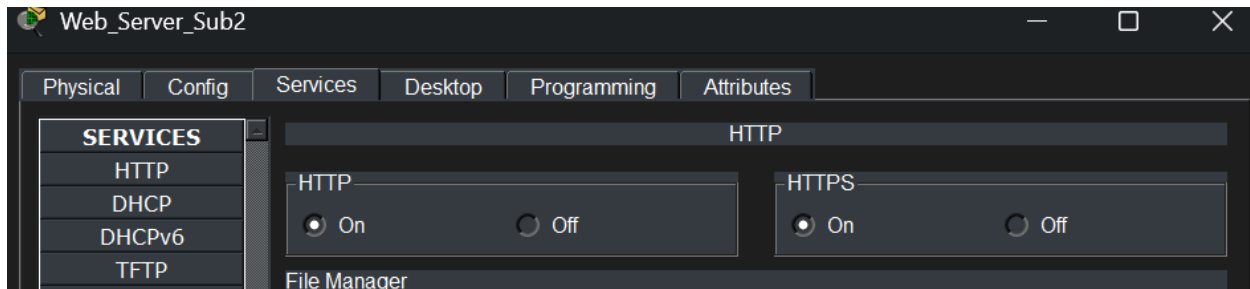
9.1 Web_Server_Main (192.168.1.5)



Web_Server_Sub_Camp_1 (192.168.2.5)

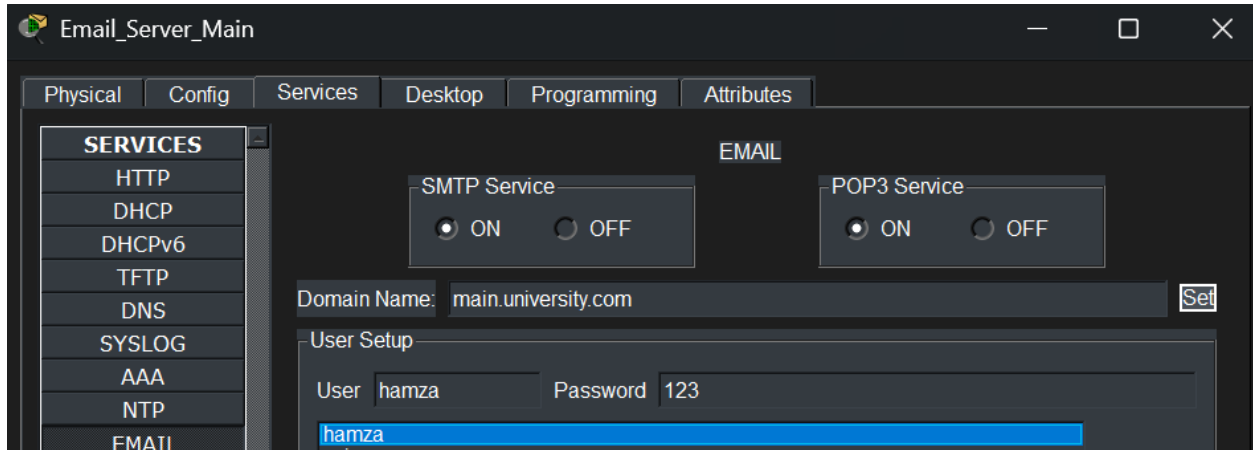


Web_Server_Sub_Camp_2 (192.168.3.5)

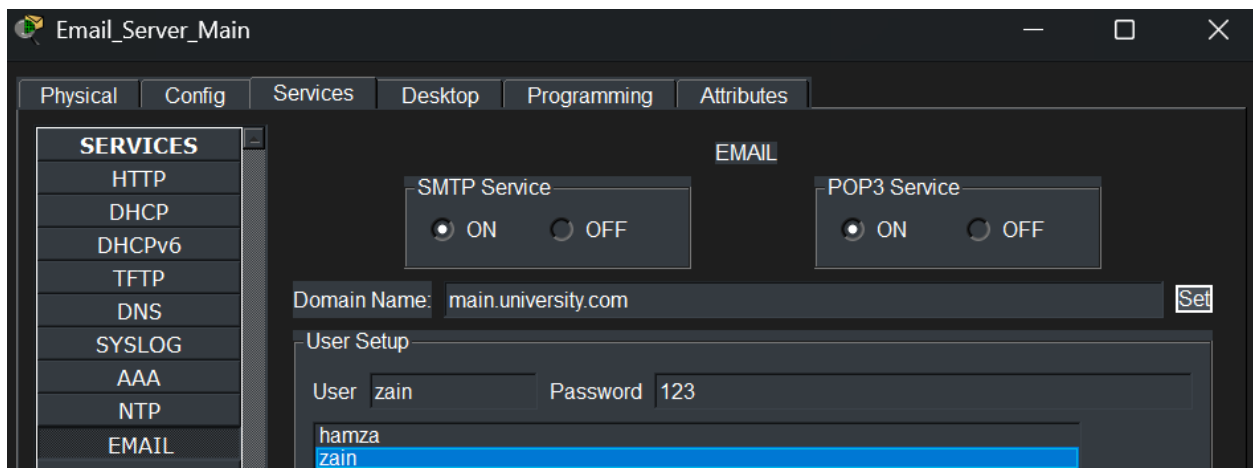


Username	Password
hamza	123
zain	123
sohaib	123

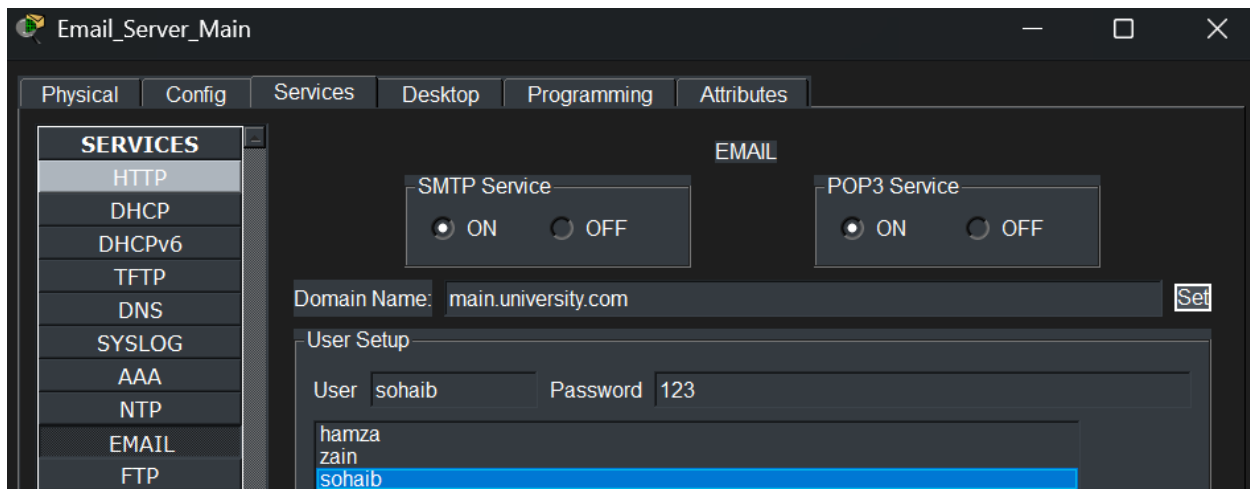
Hamza



Zain



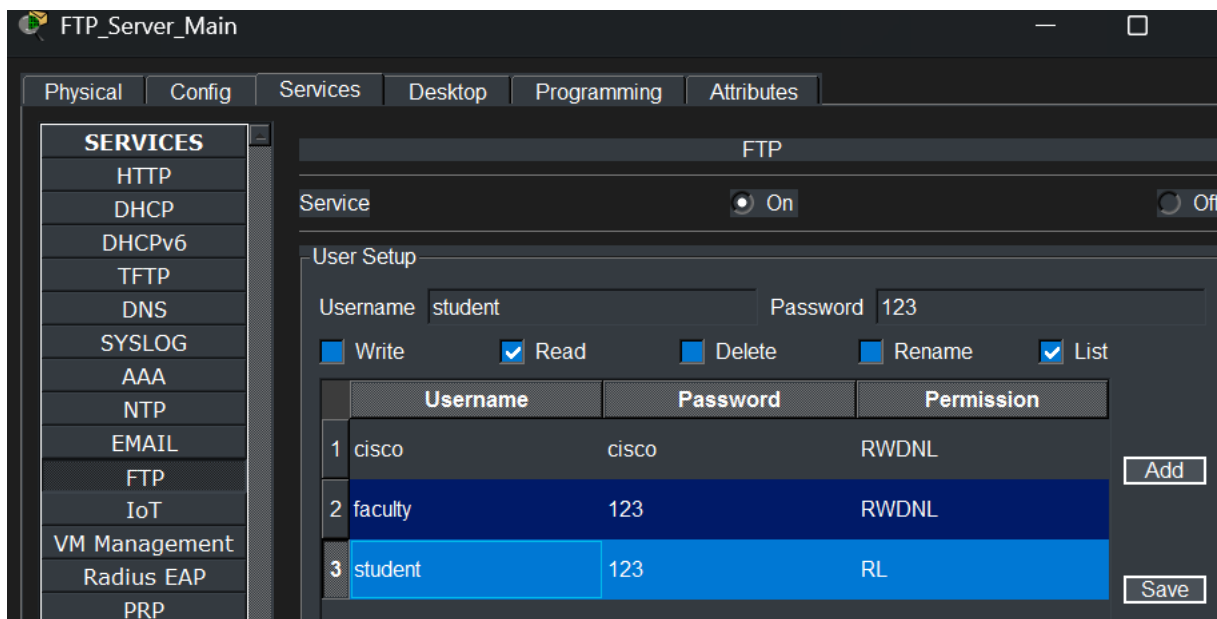
Sohaib



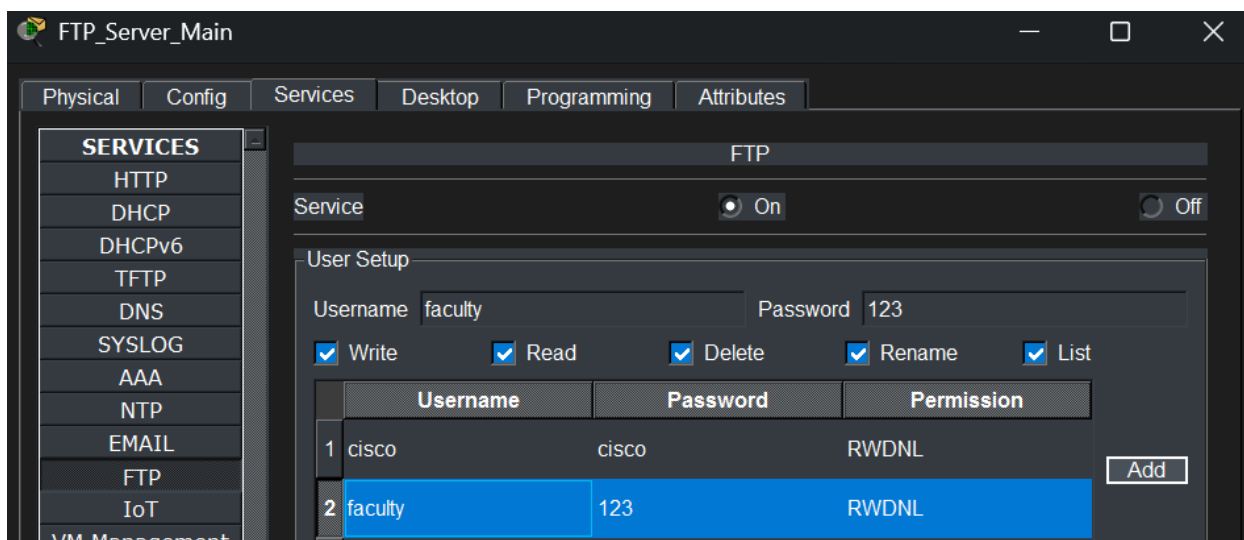
9.3 FTP_Server_Main (192.168.1.7)

Username	Password	Read	Write	Delete	Rename	List
student	123	YES	NO	NO	NO	YES
faculty	uet123	YES	YES	YES	YES	YES

Student



Faculty



10. DHCP Configuration

DHCP pools are configured on each L3 Switch for each VLAN.

Sub-Campus 1 DHCP (on L3-SW-SC1)

```
L3-SW-SC1(config)# ip dhcp excluded-address 192.168.2.1 192.168.2.10
L3-SW-SC1(config)# ip dhcp pool CS_SC1
L3-SW-SC1(dhcp-config)# network 192.168.2.0 255.255.255.224
L3-SW-SC1(dhcp-config)# default-router 192.168.2.1
L3-SW-SC1(dhcp-config)# dns-server 192.168.1.6
L3-SW-SC1(dhcp-config)# exit
```

```
L3-SW-SC1(config)# ip dhcp excluded-address 192.168.2.33 192.168.2.40
L3-SW-SC1(config)# ip dhcp pool EE_SC1
L3-SW-SC1(dhcp-config)# network 192.168.2.32 255.255.255.224
L3-SW-SC1(dhcp-config)# default-router 192.168.2.33
L3-SW-SC1(dhcp-config)# dns-server 192.168.1.6
L3-SW-SC1(dhcp-config)# exit
```

```
L3-SW-SC1(config)# ip dhcp excluded-address 192.168.2.65 192.168.2.70
L3-SW-SC1(config)# ip dhcp pool ADMIN_SC1
L3-SW-SC1(dhcp-config)# network 192.168.2.64 255.255.255.224
L3-SW-SC1(dhcp-config)# default-router 192.168.2.65
L3-SW-SC1(dhcp-config)# dns-server 192.168.1.6
L3-SW-SC1(dhcp-config)# exit
```

```
L3-SW-SC1(config)# ip dhcp excluded-address 192.168.2.97 192.168.2.97
L3-SW-SC1(config)# ip dhcp pool WIRELESS_SC1
```

```

L3-SW-SC1(dhcp-config)# network 192.168.2.96 255.255.255.224
L3-SW-SC1(dhcp-config)# default-router 192.168.2.97
L3-SW-SC1(dhcp-config)# dns-server 192.168.1.6
L3-SW-SC1(dhcp-config)# exit

```

```

L3-SW-SC1(config)# end
L3-SW-SC1# write memory

```

```

Switch#show ip dhcp binding

```

IP address	Client-ID/ Hardware address	Lease expiration	Type
192.168.2.14	0005.5E82.4843	--	Automatic
192.168.2.17	0001.63C9.BB70	--	Automatic
192.168.2.16	0030.A303.AA19	--	Automatic
192.168.2.12	00D0.BC36.E716	--	Automatic
192.168.2.13	0002.1723.3DE6	--	Automatic
192.168.2.11	0001.97E9.B5B8	--	Automatic
192.168.2.15	0030.A3A4.B482	--	Automatic
192.168.2.18	000C.CFEE.447C	--	Automatic
192.168.2.19	000B.BE0E.6C58	--	Automatic
192.168.2.20	000A.41EC.141D	--	Automatic
192.168.2.43	0001.C781.57D5	--	Automatic
192.168.2.42	000A.F340.0AA9	--	Automatic
192.168.2.41	0040.0B31.0C7C	--	Automatic
192.168.2.44	00E0.B0A4.9EC7	--	Automatic
192.168.2.45	0060.4741.0C71	--	Automatic
192.168.2.49	0001.6366.EA6A	--	Automatic
192.168.2.47	0090.2BD9.7697	--	Automatic
192.168.2.48	0060.5C2A.5AAE	--	Automatic
192.168.2.46	0090.0C09.8A40	--	Automatic
192.168.2.50	0090.2B64.6B1C	--	Automatic
192.168.2.74	0060.7029.23E4	--	Automatic
192.168.2.76	0010.1100.C985	--	Automatic
192.168.2.75	00E0.F98E.AB54	--	Automatic
192.168.2.73	00D0.97C2.1606	--	Automatic
192.168.2.72	000B.BE6C.7416	--	Automatic
192.168.2.71	00E0.A34B.D1C8	--	Automatic
192.168.2.77	0001.961C.7146	--	Automatic
192.168.2.79	0040.0BE9.7499	--	Automatic
192.168.2.78	0004.9AB2.BC7B	--	Automatic
192.168.2.80	0060.2F63.5193	--	Automatic
192.168.2.106	0002.1642.0D28	--	Automatic

```

Switch#

```

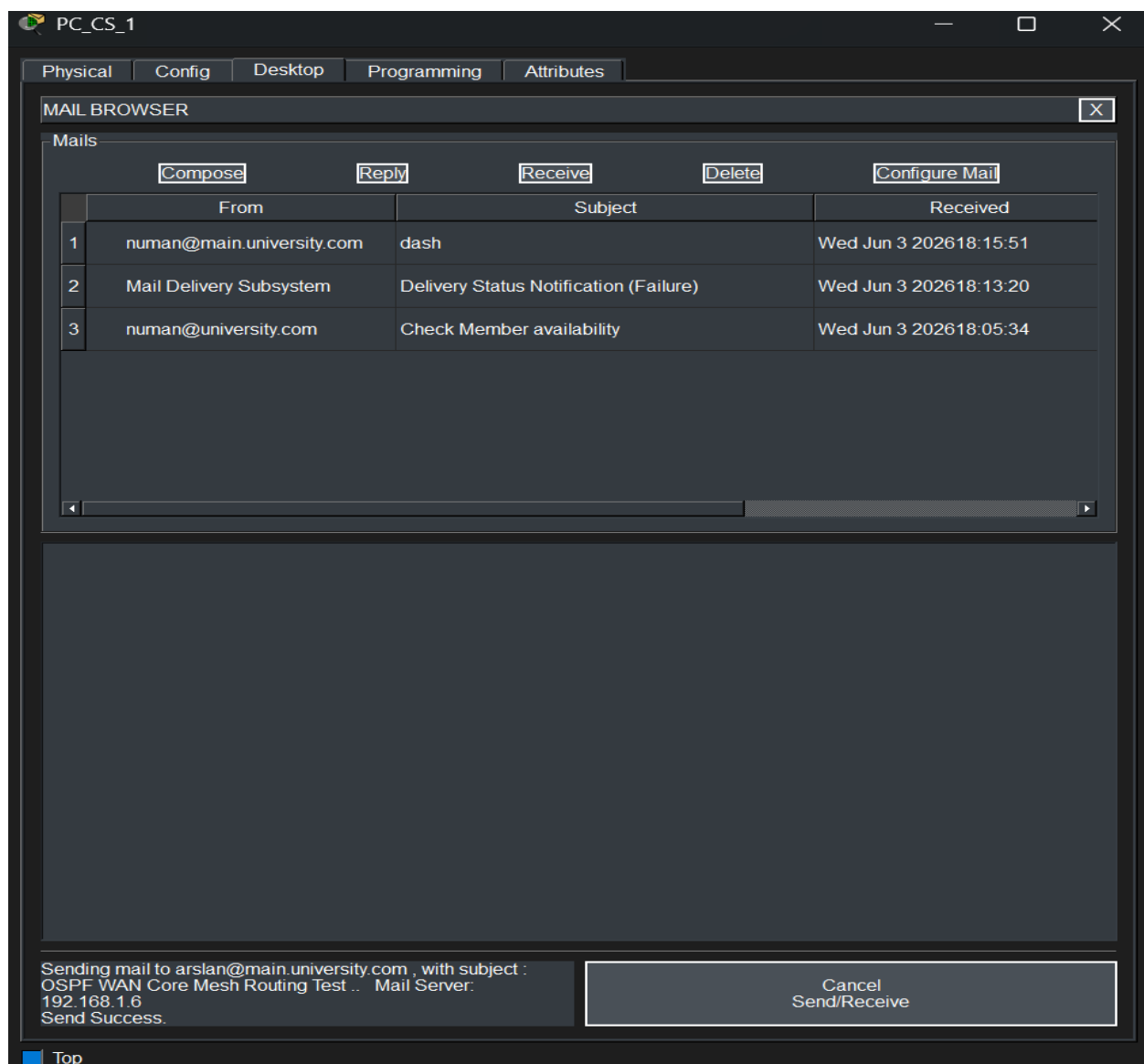
11. Email System Configuration

Email Accounts (Email_Server_Main – 192.168.1.6, Domain: university.com)

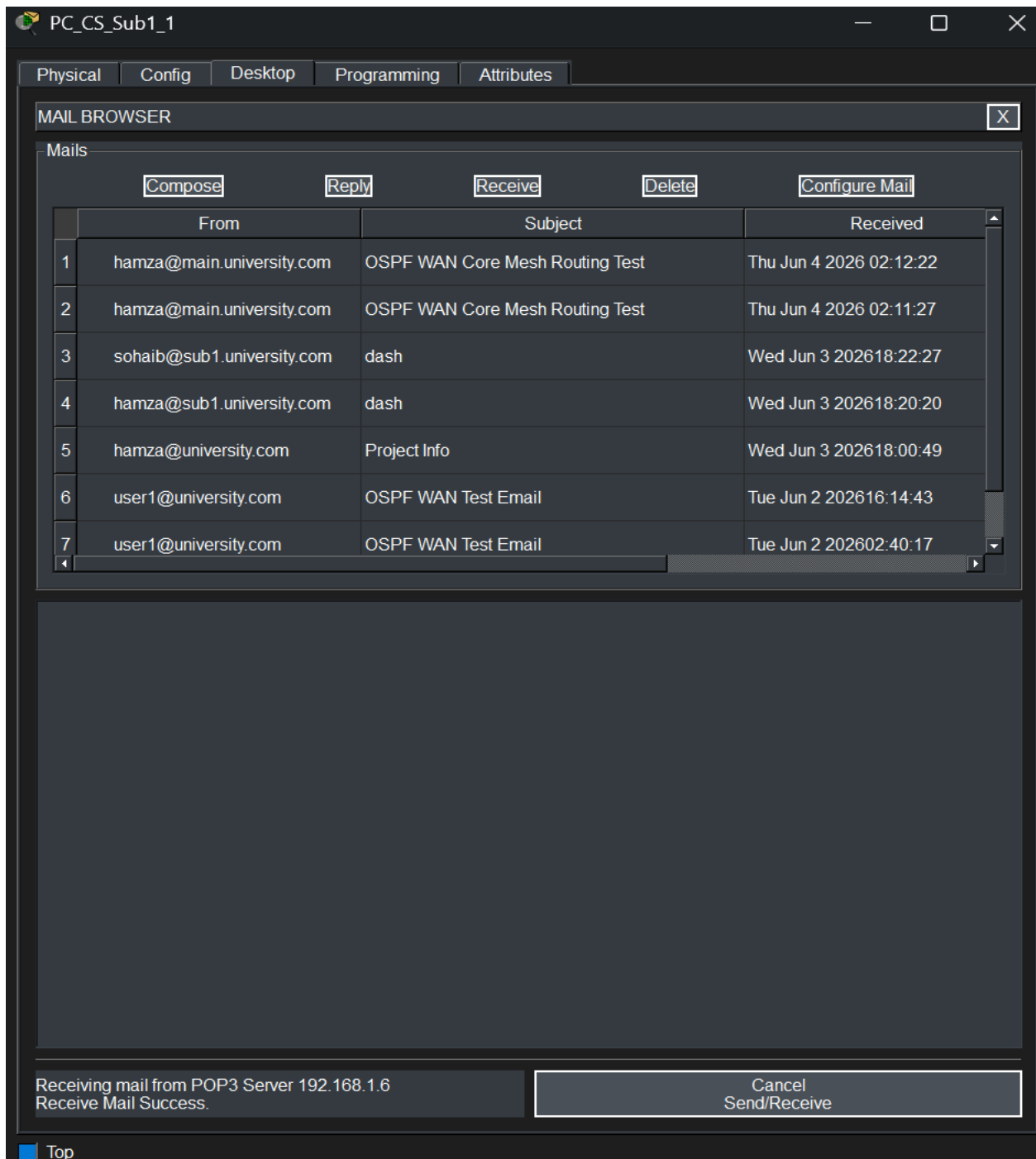
PC Name	Username	Email	Server IP
PC_CS_1 (Main)	hamza	hamza@main.university.com	192.168.1.6
PC_CS_Sub1_1 (SC1)	numan	arslan@main.university.com	192.168.1.6
PC_CS_Sub2_1 (SC2)	sohaib	sohaib@main.university.com	192.168.1.6

All clients point BOTH incoming and outgoing mail to 192.168.1.6. No DNS required.

Send Success



Receive Success



12. Test Cases & Simulations

Test 1: Intra-Department Ping (Same VLAN)

PC_CS_1 to PC_CS_2

```
C:\>ping 192.168.1.12

Pinging 192.168.1.12 with 32 bytes of data:

Reply from 192.168.1.12: bytes=32 time=1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

PC_EE_Sub1_1 to PC_EE_Sub1_2

```
C:\>ping 192.168.1.42

Pinging 192.168.1.42 with 32 bytes of data:

Reply from 192.168.1.42: bytes=32 time<1ms TTL=128
Reply from 192.168.1.42: bytes=32 time<1ms TTL=128
Reply from 192.168.1.42: bytes=32 time<1ms TTL=128
Reply from 192.168.1.42: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.42:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

PC_CS_Sub2_1 to PC_CS_Sub2_2

```
C:\>ping 192.168.2.19

Pinging 192.168.2.19 with 32 bytes of data:

Reply from 192.168.2.19: bytes=32 time<1ms TTL=128
Reply from 192.168.2.19: bytes=32 time<1ms TTL=128
Reply from 192.168.2.19: bytes=32 time<1ms TTL=128
Reply from 192.168.2.19: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.2.19:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Test 2: Inter-VLAN Ping (Same Campus)

PC_CS_1 to PC_EE_1

```
C:\>ping 192.168.1.41

Pinging 192.168.1.41 with 32 bytes of data:

Reply from 192.168.1.41: bytes=32 time=44ms TTL=127
Reply from 192.168.1.41: bytes=32 time<1ms TTL=127
Reply from 192.168.1.41: bytes=32 time<1ms TTL=127
Reply from 192.168.1.41: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.1.41:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 44ms, Average = 11ms
```

Test 3: Cross-Campus via OSPF (Main -> Sub-Campus 1)

PC_CS_1 to PC_CS_Sub1_1

Ping

```
C:\>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:

Reply from 192.168.2.1: bytes=32 time=2ms TTL=250
Reply from 192.168.2.1: bytes=32 time=1ms TTL=250
Reply from 192.168.2.1: bytes=32 time=10ms TTL=250
Reply from 192.168.2.1: bytes=32 time=1ms TTL=250

Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 10ms, Average = 3ms
```

Tracert

```
C:\>tracert 192.168.2.1

Tracing route to 192.168.2.1 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms    192.168.1.1
  1  0 ms    1 ms    0 ms    192.168.10.2
  2  0 ms    0 ms    0 ms    192.168.20.2
  3  2 ms    0 ms    1 ms    192.168.30.2
  4  1 ms    1 ms    39 ms   192.168.20.5
  5  1 ms    1 ms    10 ms   192.168.2.1

Trace complete.
```

Test 4: Web Server Access (CS Departments)

From Main Campus CS: PC_CS_1

The screenshot shows a web browser window with the address bar displaying 'http://192.168.1.5'. The page has a dark red header with the text 'UNIVERSITY OF ENGINEERING & TECHNOLOGY' and 'MAIN CAMPUS DATA CENTER • CENTRAL CLUSTER'. Below the header, there is a green box with a checkmark icon and the text '✓ Connection Node: MAIN_SERVER_CORE' and 'You have connected directly to the primary hosting farm at the Main Campus. Response times are local.' Below this, there is a white box with a lock icon and the title 'Main Cluster Core Rules'. The text below the title states 'This server hosts global student web accounts, centralized SMTP email queues, and secure FTP data repositories.' followed by two bullet points: '• Local Subnets (Main Campus): Direct communication over Gigabit infrastructure.' and '• Remote Branches (Sub-1 & Sub-2): Inbound connections routed across the high-availability OSPF WAN mesh.' Below this, there is another white box with a server icon and the title 'Infrastructure Engineering & Operations Team'. Below the title, there is a grey box with the text 'Muhammad Hamza Shahid' and 'Core Engineering Collaborators: Ahmad • Arslan • Sohaib • Zain • Hassan • Amar • Daood'. At the bottom of the page, there is a dark red footer with the text '© 2026 UET Main Cluster Central Server Infrastructure.'

Web Browser

URL: http://192.168.1.5

Go Stop

UNIVERSITY OF ENGINEERING & TECHNOLOGY

MAIN CAMPUS DATA CENTER • CENTRAL CLUSTER

✓ **Connection Node: MAIN_SERVER_CORE**

You have connected directly to the primary hosting farm at the Main Campus. Response times are local.

🔒 Main Cluster Core Rules

This server hosts global student web accounts, centralized SMTP email queues, and secure FTP data repositories.

- **Local Subnets (Main Campus):** Direct communication over Gigabit infrastructure.
- **Remote Branches (Sub-1 & Sub-2):** Inbound connections routed across the high-availability OSPF WAN mesh.

🖥️ Infrastructure Engineering & Operations Team

Muhammad Hamza Shahid

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© 2026 UET Main Cluster Central Server Infrastructure.

From Sub-Campus 1 CS: PC_CS_Sub1_1

Web Browser

< > URL http://192.168.2.5

Go Stop

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SUB-CAMPUS 1 NODE • REGIONAL MIRROR

✓ **Connection Node: SUB_CAMPUS_1_LOCAL**

Traffic successfully processed by the Sub-Campus 1 distribution layer switch before rendering.

 **Sub-Campus 1 Operating Matrix**

This mirror repository offloads local caching queries for Sub-Campus 1 computer labs, department workstations, and wireless devices.

- **VLAN 30 Security Check:** Port 80 packets from local Admin PCs are explicitly intercepted and dropped at this gateway.

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© 2026 UET Sub-Campus 1 Local Operations Cache.

From Sub-Campus 2 CS: Open PC_CS_Sub2_1

Web Browser

< > URL http://192.168.3.5

Go Stop

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SUB-CAMPUS 2 NODE • EDGE REPOSITORY

✓ Connection Node: SUB_CAMPUS_2_EDGE

Data stream processed over the Sub-Campus 2 multi-point serial links and edge routing framework.

 Sub-Campus 2 Path Architecture

This server handles localized intra-VLAN academic routing loops and hosts local digital content matrices for Sub-Campus 2 departments.

- **Backup Routing Path Active:** Fully integrated with the newly installed redundant 4th border router core for bulletproof link failover.

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© 2026 UET Sub-Campus 2 Redundant Edge Node.

Test 5: FTP Access

Login:

Username: student

Password: 123

dir

DIR

```
C:\>dir

Volume in drive C has no label.
Volume Serial Number is 5E12-4AF3
Directory of C:\

1/1/1970    5:0 PM                47      information.txt
1/1/1970    5:0 PM                52      project.txt
1/1/1970    5:0 PM                26      sampleFile.txt
               125 bytes              3 File(s)
```

Test 6: Troubleshooting Commands

Complete Command Reference

Command	Run On	Verifies
show ip route	All Routers & L3 SW	Routing table – O/B/C entries
show ip ospf neighbor	Routers + L3 SW	OSPF adjacency FULL state
show ip bgp summary	Border Routers	BGP sessions established
show vlan brief	Access Switches	VLAN active, ports assigned
show interfaces trunk	L3 Switches	Trunk links + allowed VLANs
show ip dhcp binding	L3 Switches	DHCP leases assigned
show access-lists	Routers + L3 SW	ACL hit counters
show running-config	Any device	Full running configuration
ping X.X.X.X	PCs	Layer 3 connectivity
tracert X.X.X.X	PCs	Hop-by-hop routing path
ftp X.X.X.X	PCs	FTP server reachability

Test 7: Redundancy Check

Before Shutdown on Main Border Router

```
Router>enable
Router#traceroute 192.168.3.21
Type escape sequence to abort.
Tracing the route to 192.168.3.21

 1  105.12.32.1      0 msec    0 msec    0 msec
 2  105.12.32.1      !H         *        !H
 3  *                *

Router#
```

After Shutdown on Main Border (Totally Different Path)

```
Router#traceroute 192.168.3.21
Type escape sequence to abort.
Tracing the route to 192.168.3.21

 1  192.168.30.9     0 msec    1 msec    2 msec
 2  192.168.20.9     19 msec   2 msec    16 msec
 3  192.168.10.9     14 msec   1 msec    2 msec
 4  *                0 msec    0 msec
```

13. Conclusion

Final Checklist

Requirement	Implementation	Status
8 Routers total	ISP + BR-Main + BR-SC1 + BR-SC2 + IR-Main + IR-SC1 + IR-SC2 + IR-Server	Done
3 L3 Switches	L3-Core-SW, L3-SW-SC1, L3-SW-SC2	Done
3 Access Switches	SW-Main, SW-SC1, SW-SC2	Done
VLAN isolation (10/20/30)	Configured on all switches	Done
Wireless VLANs (40/41/42)	Access Points per campus	Done
/27 VLSM subnetting	All 12 department subnets	Done
OSPF internal routing	All routers FULL adjacency	Done
BGP inter-campus routing	WAN triangle + ISP peering	Done
Web Server 192.168.1.5	HTTP/HTTPS enabled	Done
Email Server 192.168.1.6	university.com, 3 users	Done
FTP Server 192.168.1.7	student + faculty accounts	Done

DHCP all departments	Pools on L3 switches	Done
ACL – Admin blocked Port 80	All 3 campuses Admin subnets	Done
ACL – Admin email permitted	SMTP/POP3 allowed	Done
Firewall on border routers	External inbound blocked	Done
OSPF failover tested	1-2 packet loss on failover	Done
Simulation mode ACL proof	Red X at switch boundary	Done
Full show command audit	All devices verified	Done